Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

801

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = −x + 7

GRADING NOTE Any equivalent equation. Midpoint (4, 3); slope of AB is 1, so the bisector has slope −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

802

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(7, 0)

GRADING NOTE Form required: an ordered pair. It is where y = −x + 7 meets y = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

803

QUESTION ID

S19-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Distances are $\sqrt{32} \approx 5.66$ and $\sqrt{20} \approx 4.47$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

804

QUESTION ID

S19-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 7)

GRADING NOTE Points on x = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

805

QUESTION ID

S19-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The vertical line x = 1.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

807

QUESTION ID

S19-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three perpendicular bisectors meet at O, and OA = OB = OC.

GRADING NOTE REQUIRED WORK: Let the bisectors of AB and BC meet at O; use the theorem and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of AB and BC are not parallel (AB and BC are not parallel), so they meet at O; OA = OB and OB = OC (theorem); so OA = OC; by the converse O is on the bisector of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

808

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 4)

GRADING NOTE Form required: an ordered pair. Bisectors x = 3 and y = 4 meet at (3, 4); it is 5 from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

809

QUESTION ID

S19-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For an obtuse triangle it is outside; for a right triangle it is on the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns that are not on one straight line will share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

810

QUESTION ID

S19-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Where the perpendicular bisectors of the triangle's sides meet

GRADING NOTE Points equidistant from two towns are on the perpendicular bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

811

QUESTION ID

S19-F2-Q5

TYPE

Short answer

CORRECT ANSWER

OA = OB and OB = OC give OA = OC, and by the converse O is on the perpendicular bisector of AC.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

SET 20 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

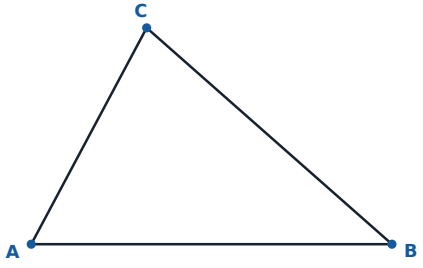
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
m∠A + m∠B + m∠C = 180° in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

814

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠C = 180°

GRADING NOTE REQUIRED WORK: Name the two new angles at C, use alternate interior angles with AB, and the straight angle at C.

Any valid proof (paragraph, two-column, or flow). Essential: (1) through C there is exactly one line parallel to AB (parallel postulate); (2) the angles it makes with CA and CB are congruent to ∠A and ∠B (alternate interior angles, transversals CA and CB); (3) those two angles and ∠ACB form a straight angle, 180°; (4) substitute. Invalid: measuring angles; citing the exterior angle theorem, because in this course that theorem is proved FROM the angle sum (set 20, FORWARD 1), so using it here would be circular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

815

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

m∠P + m∠Q + m∠R = 180°

GRADING NOTE REQUIRED WORK: Same strategy with vertex P: alternate interior angles and a straight angle at P.

Any valid proof (paragraph, two-column, or flow). Essential: unique parallel through P; the two angles at P next to ∠QPR are congruent to ∠Q and ∠R (alternate interior); the three angles at P make 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

816

QUESTION ID

S20-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠B + m∠ACB = 180°

GRADING NOTE REQUIRED WORK: Use alternate interior angles (∠A and ∠ACE) and corresponding angles (∠B and ∠ECD), then the straight angle BCD.

Any valid proof (paragraph, two-column, or flow). Essential: ∠ACE ≅ ∠A (alternate interior, transversal AC); ∠ECD ≅ ∠B (corresponding, transversal BD); ∠ACB + ∠ACE + ∠ECD = 180° (straight angle at C).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

In triangle ABC, translate the triangle along ray AB so that A moves to B; C moves to C′ and B moves to D (on ray AB beyond B). Prove that the angles of triangle ABC add to 180°, using the three angles at B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

817

QUESTION ID

S20-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

m∠A + m∠ABC + m∠C = 180°

GRADING NOTE REQUIRED WORK: Use the straight angle ABD at B, the translation (∠C′BD ≅ ∠A), and alternate interior angles (BC′ ∥ AC).

Any valid proof (paragraph, two-column, or flow). Essential: ∠ABC + ∠CBC′ + ∠C′BD = 180° (straight angle, A, B, D collinear); ∠C′BD ≅ ∠CAB (the translation maps ∠CAB onto it); the translation maps line AC to the parallel line BC′, so ∠CBC′ ≅ ∠ACB (alternate interior angles, transversal BC); substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

In triangle PQR, extend side PQ past Q to S, and draw ray QT parallel to PR (T on the same side of line PS as R). Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

818

QUESTION ID

S20-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

m∠P + m∠PQR + m∠R = 180°

GRADING NOTE REQUIRED WORK: Use alternate interior angles (∠R and ∠RQT) and corresponding angles (∠P and ∠TQS), then the straight angle PQS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠RQT ≅ ∠R (alternate interior, transversal QR); ∠TQS ≅ ∠P (corresponding, transversal PS); ∠PQR + ∠RQT + ∠TQS = 180° (straight angle at Q); substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

820

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

821

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

822

QUESTION ID

S20-B1-Q3

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form. 6x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

823

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Its alternate interior angle

GRADING NOTE Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

824

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two angles over 90° would already add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

826

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

827

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They make a straight angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

∠1, ∠2, and ∠3 form a straight angle, with ∠1 = ∠3 = 50°. Find ∠2.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

828

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 80°

GRADING NOTE 180 − 100 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

829

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form. 5x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

830

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 180° but can be any pair, such as 30° and 150°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

832

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

58°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

833

QUESTION ID

S20-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the theorem proved in an earlier set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Triangle ABC has A at the left end of base AB, B at the right end, and C above AB. Line ℓ through C is parallel to AB; D is on ℓ to the left of C and E is on ℓ to the right of C. Which angle is congruent to ∠CAB?

A. ∠ACB

B. ∠ECA

C. ∠DCA

D. ∠ECB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

834

QUESTION ID

S20-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. ∠DCA

GRADING NOTE With transversal CA, ∠CAB and ∠DCA are alternate interior angles (on opposite sides of CA, between the parallel lines). ∠ECB is congruent to ∠CBA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

835

QUESTION ID

S20-B3-Q4

TYPE

Short answer

CORRECT ANSWER

The parallel postulate: through a point not on a line there is exactly one parallel line.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

836

QUESTION ID
S20-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only when the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that m∠ACD = m∠A + m∠B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

838

QUESTION ID

S20-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

m∠ACD = m∠A + m∠B

GRADING NOTE REQUIRED WORK: Use the linear pair at C and the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: m∠ACD + m∠ACB = 180° (linear pair); m∠A + m∠B + m∠ACB = 180° (angle sum); substitute and subtract m∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

839

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Remote interior: 77°; adjacent interior: 55°.

GRADING NOTE Form required: both angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

840

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exterior angle at an obtuse vertex is acute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

841

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 25

GRADING NOTE 5x + 10 = 4x + 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

842

QUESTION ID
S20-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Each exterior angle is 180° minus its interior angle, so the sum is 540° − 180° = 360°.

GRADING NOTE Any valid argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

844

QUESTION ID

S20-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

$(n - 2) \cdot 180°$

GRADING NOTE REQUIRED WORK: Draw the diagonals from one vertex to divide the polygon into triangles and use the triangle angle sum.

Any valid proof (paragraph, two-column, or flow). Essential: from one vertex there are n − 3 diagonals, making n − 2 triangles; their angles together make up exactly the polygon's angles; each triangle contributes 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

845

QUESTION ID
S20-F2-Q2

TYPE
Short answer

CORRECT ANSWER
135°

GRADING NOTE Any equivalent form. $(8 - 2) \cdot 180 \div 8 = 135$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

846

QUESTION ID

S20-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. Each exterior angle is 24°; 360 ÷ 24 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

847

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(6 - 2) \cdot 180 = 720°$; 1,080° is the sum for an octagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

848

QUESTION ID

S20-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 115

GRADING NOTE 540 − 425 = 115.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

SET 21 • HSG-CO.C.10

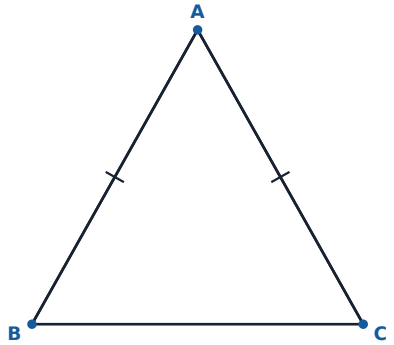
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

851

QUESTION ID

S21-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE REQUIRED WORK: Prove △ABD ≅ △ACD by SAS, then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC (given); ∠BAD ≅ ∠CAD (definition of angle bisector); AD = AD; △ABD ≅ △ACD (SAS); ∠B ≅ ∠C (corresponding angles). Invalid: "∠B and ∠C look equal"; using the fact that AD ⊥ BC (not yet known).

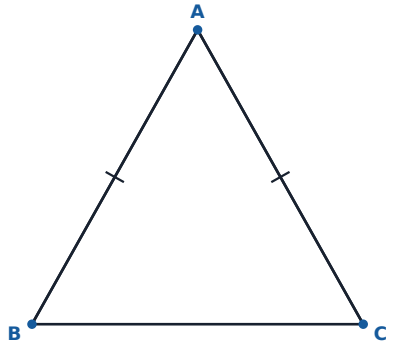
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

852

QUESTION ID

S21-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠B ≅ ∠C

GRADING NOTE

REQUIRED WORK: Prove △ABM ≅ △ACM by SSS.

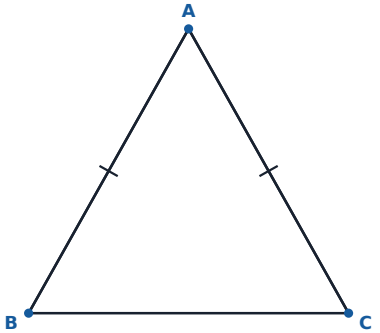
Any valid proof (paragraph, two-column, or flow). Essential: AB = AC; BM = CM (midpoint); AM = AM; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

853

QUESTION ID

S21-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The reflection maps ∠ABC onto ∠ACB, so ∠B ≅ ∠C.

GRADING NOTE REQUIRED WORK: Explain where the reflection sends ray AB, point B, and point C.

Any valid explanation. Essential: the reflection across the bisector of ∠A swaps rays AB and AC (the bisector makes equal angles with them); B goes to the point of ray AC at distance AB = AC, which is C; so B ↔ C and the triangle maps onto itself with ∠B ↔ ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

854

QUESTION ID

S21-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

All three angles are congruent (each is 60°).

GRADING NOTE REQUIRED WORK: Apply the isosceles base angle theorem twice, with two different pairs of equal sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AC gives ∠B ≅ ∠C; AB = BC gives ∠A ≅ ∠C; so all three are congruent (and each is 60° by the angle sum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

Triangle ABC has A(0, h), B(−b, 0), and C(b, 0), with b > 0 and h > 0, so AB = AC. Prove that ∠B ≅ ∠C using the reflection (x, y) → (−x, y).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

855

QUESTION ID

S21-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The reflection fixes A and swaps B and C, so it maps ∠ABC onto ∠ACB; reflections keep angle measure.

GRADING NOTE REQUIRED WORK: Find the images of A, B, and C, and explain why the reflection carries ∠ABC onto ∠ACB.

Any valid explanation. Essential: A(0, h) → A, B(−b, 0) → C(b, 0), C → B; so ray BA → ray CA and ray BC → ray CB; ∠ABC → ∠ACB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

857

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM.
Which criterion shows △ABM ≅ △ACM?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

858

QUESTION ID
S21-B1-Q2

TYPE
Short answer

CORRECT ANSWER
SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

859

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

860

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Two sides and a non-included angle

GRADING NOTE SSA is not a criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If △ABD ≅ △ACD, then ∠ABD ≅ ∠ACD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

861

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

863

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

865

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

866

QUESTION ID

S21-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. m∠BAD = m∠DAC

GRADING NOTE Equal angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of ∠A of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

867

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only when AB = AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

869

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Base QR; base angles ∠Q and ∠R

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

870

QUESTION ID

S21-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition used in this course.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

871

QUESTION ID

S21-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Sides 4, 5, 6

GRADING NOTE No two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

872

QUESTION ID

S21-B3-Q4

TYPE

Short answer

CORRECT ANSWER

∠X

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

873

QUESTION ID

S21-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It has at least two congruent sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, ∠B ≅ ∠C. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

875

QUESTION ID

S21-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = AC

GRADING NOTE REQUIRED WORK: Draw the bisector of ∠A meeting BC at D and prove △ABD ≅ △ACD by AAS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (given); ∠BAD ≅ ∠CAD (bisector); AD = AD; AAS; AB = AC. Invalid: using AB = AC to choose the auxiliary line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

876

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4; PQ = 14

GRADING NOTE Form required: x and PQ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

877

QUESTION ID

S21-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the converse of the isosceles triangle theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has m∠A = 50° and m∠B = 65°. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

878

QUESTION ID

S21-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. AB and AC

GRADING NOTE m∠C = 65° = m∠B, so the sides opposite them, AC and AB, are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

879

QUESTION ID

S21-F1-Q5

TYPE

Answer with required work

CORRECT ANSWER

All three sides are congruent.

GRADING NOTE

REQUIRED WORK: Apply the converse twice.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A ≅ ∠B gives BC = AC; ∠B ≅ ∠C gives AC = AB; so all three sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

881

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

71°

GRADING NOTE Any equivalent form. (180 − 38) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

882

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. Triangles ABD and ACD are isosceles, so ∠BAD = ∠B and ∠CAD = ∠C; then 2(∠B + ∠C) = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

883

QUESTION ID

S21-F2-Q3

TYPE

Answer with required work

CORRECT ANSWER

AD = AE

GRADING NOTE REQUIRED WORK: Use the base angles and SAS on triangles ABD and ACE.

Any valid proof (paragraph, two-column, or flow). Essential: ∠B ≅ ∠C (base angles); AB = AC; BD = CE; △ABD ≅ △ACE (SAS); AD = AE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

884

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (180 − 100) ÷ 2 = 40°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

885

QUESTION ID

S21-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 44°

GRADING NOTE ∠PQR = 68° = ∠R, so ∠P = 180 − 136 = 44.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

888

QUESTION ID

S22-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

D(b, c), E(a + b, c); DE is horizontal like AB, and DE = a = ½(2a) = ½AB.

GRADING NOTE REQUIRED WORK: Find D and E with the midpoint formula, then compare slopes and lengths.

Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; DE has slope 0 (same y) and AB has slope 0, so DE ∥ AB (and DE ≠ AB since c ≠ 0); DE = |a + b − b| = a and AB = 2a. Invalid: checking one numerical triangle only.

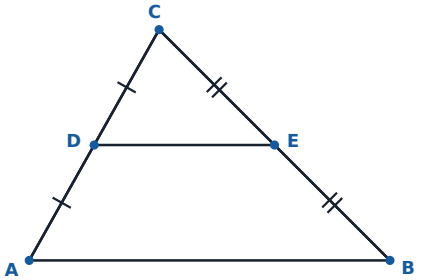
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

889

QUESTION ID

S22-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

DE ∥ AB and DE = ½AB

GRADING NOTE REQUIRED WORK: Prove △DEC ≅ △FEB (SAS), deduce BF = DC = AD and BF ∥ AD, so ABFD is a parallelogram; then use its opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: CE = BE, DE = FE, vertical angles at E, so △DEC ≅ △FEB (SAS); BF = DC = AD; ∠DCE ≅ ∠FBE gives BF ∥ AC (alternate interior converse); one pair of opposite sides of ABFD is parallel and congruent, so ABFD is a parallelogram; so DF ∥ AB and DF = AB; DE = ½DF = ½AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. F is the midpoint of AB and E is the midpoint of AC. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

890

QUESTION ID

S22-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

F(a, 0), E(b, c); FE goes from (a, 0) to (b, c), the same direction as BC from (2a, 0) to (2b, 2c), and FE = ½BC.

GRADING NOTE REQUIRED WORK: Find F and E, then compare slopes (or note both are vertical) and lengths.

Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; FE = (b − a, c) and BC = (2b − 2a, 2c) = 2(b − a, c), so they are parallel (equal slopes c/(b − a), or both vertical when b = a) and FE = √((b − a)² + c²) = ½BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

In triangle ABC, F is the midpoint of AB and E is the midpoint of AC. Extend FE past E to G so that EG = FE. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

891

QUESTION ID

S22-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

FE ∥ BC and FE = ½BC

GRADING NOTE REQUIRED WORK: Prove △AEF ≅ △CEG (SAS); then show FBCG is a parallelogram.

Any valid proof (paragraph, two-column, or flow). Essential: AE = CE, FE = GE, vertical angles at E, so △AEF ≅ △CEG (SAS); CG = AF = FB and ∠FAE ≅ ∠GCE, so CG ∥ AB (alternate interior converse); FBCG has a pair of opposite sides parallel and congruent, so it is a parallelogram; FG ∥ BC and FG = BC; FE = ½FG = ½BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(2p, 2q), B(2r, 2s), and C(2t, 2u), not on one line, with AB not vertical. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

892

QUESTION ID

S22-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

D(p + t, q + u), E(r + t, s + u); DE = (r − p, s − q) and AB = (2r − 2p, 2s − 2q) = 2·DE, so DE ∥ AB and DE = ½AB.

GRADING NOTE REQUIRED WORK: Find D and E, then compare the coordinate differences of DE and AB.

Any valid proof (paragraph, two-column, or flow). Essential: both midpoints; slope DE = (s − q)/(r − p) = slope AB; DE = √((r − p)² + (s − q)²) = ½AB. DE and AB are different lines because C is not on AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

894

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

GRADING NOTE Form required: an ordered pair. Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (−3, 1) to (5, −7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

895

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(1, −3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

896

QUESTION ID

S22-B1-Q3

TYPE

Short answer

CORRECT ANSWER

(a, c)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

897

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE ((1 + 7) ÷ 2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

898

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is (3, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

900

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. (8 − 2) ÷ (4 − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (b, c) and (a + b, c), where a ≠ 0.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

901

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form. The y-coordinates are equal and the points are different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

902

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same steepness, different positions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

903

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. From (1, 3) to (5, 5)

GRADING NOTE Both have slope 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

904

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Horizontal has slope 0; vertical is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

906

QUESTION ID

S22-B3-Q1

TYPE

Short answer

CORRECT ANSWER

a (that is, |a|, with a > 0)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

907

QUESTION ID

S22-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from (−2, 5) to (4, 5).

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

908

QUESTION ID

S22-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

GRADING NOTE Horizontal: |4 − (−2)| = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

909

QUESTION ID

S22-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{(9 + 16)} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when a > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

910

QUESTION ID

S22-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

912

QUESTION ID

S22-B4-Q1

TYPE

Short answer

CORRECT ANSWER

$\ell \parallel m$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

913

QUESTION ID
S22-B4-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Converse of the alternate interior angles theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In △ABC, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that EF = DE, and △DEC ≅ △FEB. Which fact shows BF ∥ AC?

A. DE = EF

B. ∠DCE ≅ ∠FBE are alternate interior angles

C. BF = DC

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

914

QUESTION ID

S22-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. ∠DCE ≅ ∠FBE are alternate interior angles

GRADING NOTE Congruent alternate interior angles (transversal BC) give parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

915

QUESTION ID

S22-B4-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

916

QUESTION ID

S22-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Vertical angles are congruent for any crossing lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

918

QUESTION ID

S22-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ ∥ AC ∥ SR and PQ = ½AC = SR, so PQRS is a parallelogram.

GRADING NOTE REQUIRED WORK: Draw diagonal AC and apply the midsegment theorem in triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: in △ABC, PQ ∥ AC and PQ = ½AC; in △ADC, SR ∥ AC and SR = ½AC; so PQ ∥ SR and PQ = SR; a quadrilateral with one pair of opposite sides parallel and congruent is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

919

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any equivalent form. Its sides are ½AC, ½BD, ½AC, ½BD: 5 + 7 + 5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

920

QUESTION ID

S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is Varignon's theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

921

QUESTION ID

S22-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. rectangle

GRADING NOTE Its sides are parallel to the diagonals, so adjacent sides are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

922

QUESTION ID

S22-F1-Q5

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. Each midsegment is half the parallel side: 6 + 8 + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

924

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each side of DEF is half of a side of ABC (midsegment theorem), so the triangles are similar by SSS similarity with scale factor ½.

GRADING NOTE Any equivalent explanation with the factor ½ (or 2 from DEF to ABC).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

925

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form. Scale factor ½ gives area factor ¼; also the 4 small triangles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

True or false?

The four triangles formed by drawing all three midsegments of a triangle are similar to it with scale factor ⅓.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

926

QUESTION ID
S22-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The scale factor is ½: each side is half a side of the original.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

A. 9

B. 18

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

927

QUESTION ID

S22-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.5

GRADING NOTE Each repetition halves the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

928

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(4, 0), (5, 3), (1, 3); area of ABC = 24 and area of the midsegment triangle = 6, a ratio of 1 : 4.

GRADING NOTE Form required: the three midpoints and both areas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

Prove that the medians of a triangle meet at one point

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints, including with variables	**6.G.A.3**
BACKWARD 2	Use the midsegment theorem	**HSG-CO.C.10**
BACKWARD 3	Find the point a given fraction of the way along a segment	**7.RP.A.2**
BACKWARD 4	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Locate the centroid and use the 2 : 1 ratio	**HSG-GPE.B.6**
FORWARD 2	Prove that the angle bisectors of a triangle meet at one point	**HSG-C.A.3**

SET 23 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

931

QUESTION ID

S23-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

G is on the median from A (to (b + c, d)), from B (to (c, d)), and from C (to (b, 0)), so the medians meet at G.

GRADING NOTE REQUIRED WORK: Find the midpoint of each side, then show G is two-thirds of the way from each vertex to the opposite midpoint (or on each median line).

Any valid proof (paragraph, two-column, or flow). Essential: the three midpoints (b + c, d), (c, d), (b, 0); for each median, G = vertex + ⅔(midpoint − vertex): A + ⅔((b + c, d) − A) = G; B + ⅔((c, d) − (2b, 0)) = G; C + ⅔((b, 0) − (2c, 2d)) = G. Alternative: show G satisfies each median's equation. Invalid: checking one triangle with numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

932

QUESTION ID

S23-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

BG = 2GE; the same argument for medians AD and BE gives the point that is two-thirds along BE, which is G, so AD passes through G.

GRADING NOTE REQUIRED WORK: Use the midsegment theorem in triangles ABC and GBC to show FEKH is a parallelogram, then its diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: FE ∥ BC and FE = ½BC (midsegment in △ABC); HK ∥ BC and HK = ½BC (midsegment in △GBC); so FEKH is a parallelogram; its diagonals bisect each other, so HG = GE; with BH = HG, BG = 2GE; any two medians meet at the point two-thirds along each, so all three share it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle ABC has A(−2a, 0), B(2a, 0), and C(0, 2b), with a > 0 and b > 0. Prove that the three medians meet at (0, 2b/3).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

933

QUESTION ID

S23-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Midpoints (a, b), (−a, b), (0, 0); the point (0, 2b/3) is ⅔ of the way along each median.

GRADING NOTE REQUIRED WORK: Find the three midpoints and show that (0, 2b/3) is two-thirds of the way from each vertex to the opposite midpoint.

Any valid proof (paragraph, two-column, or flow). Essential: midpoint of BC (a, b), of AC (−a, b), of AB (0, 0); A + ⅔((a, b) − A) = (0, 2b/3); B + ⅔((−a, b) − B) = (0, 2b/3); C + ⅔((0, 0) − C) = (0, 2b/3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

934

QUESTION ID

S23-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The medians meet at (2, 2).

GRADING NOTE REQUIRED WORK: Find each median as a line through a vertex and a midpoint, find where two meet, and check that the third passes through that point.

Any valid proof (paragraph, two-column, or flow). Essential: midpoints (3, 3), (0, 3), (3, 0); medians y = x, y = −½x + 3 (from Q), y = −2x + 6 (from R); y = x and y = −2x + 6 meet at (2, 2); (2, 2) satisfies y = −½x + 3. (This proves the claim for this triangle; the general proof needs variables.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

Triangle ABC has $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, not on one line. Let $G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$. Prove that G lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

935

QUESTION ID

S23-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

For the median from A: $A + \frac{2}{3}(((x_2 + x_3)/2, (y_2 + y_3)/2) - A) = G$; the same holds from B and C, so the medians meet at G.

GRADING NOTE REQUIRED WORK: For each median, show G = vertex + ⅔(opposite midpoint − vertex).

Any valid proof (paragraph, two-column, or flow). Essential: the computation for one median with the general coordinates and the symmetry (or computation) for the other two. Invalid: drawing examples.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

937

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(b + c, d)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

938

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 3)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

A. the midpoint of the opposite side

B. another vertex

C. the foot of the perpendicular from that vertex

D. the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

939

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the midpoint of the opposite side

GRADING NOTE Definition of a median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

A triangle has exactly three medians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

940

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One from each vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

A median always bisects the angle at its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

941

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in special cases (for example, the vertex angle of an isosceles triangle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

943

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

944

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Midsegment theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

A. HK ⊥ BC

B. HK ∥ GB

C. HK ∥ BC and HK = ½BC

D. HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

945

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. HK ∥ BC and HK = ½BC

GRADING NOTE Midsegment theorem in triangle GBC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

946

QUESTION ID

S23-B2-Q4

TYPE

Short answer

CORRECT ANSWER

22 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

True or false?

A midsegment of a triangle is always half as long as each of the other two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

947

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only half as long as the side it is parallel to.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

949

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

950

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. 3 + ⅔(9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

A. 2 : 3

B. 3 : 2

C. 1 : 2

D. 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

951

QUESTION ID

S23-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 : 1

GRADING NOTE ⅔ and ⅓ of the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

True or false?

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

952

QUESTION ID
S23-B3-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE ⅔ of each coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

True or false?

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

953

QUESTION ID
S23-B3-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

955

QUESTION ID

S23-B4-Q1

TYPE

Short answer

CORRECT ANSWER

y = x

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

956

QUESTION ID

S23-B4-Q2

TYPE

Short answer

CORRECT ANSWER

$y = -\frac{1}{2}x + 3$

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

Which point is on the line $y = -2x + 6$?

A. (1, 2)

B. (3, 3)

C. (0, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

957

QUESTION ID

S23-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 2)

GRADING NOTE $-2(2) + 6 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

True or false?

The line through (0, 6) and (3, 0) has slope −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

958

QUESTION ID

S23-B4-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (0 − 6) ÷ (3 − 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

True or false?

The line y = x passes through (2, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

959

QUESTION ID

S23-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2 \neq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

961

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 5)

GRADING NOTE Form required: an ordered pair. Average the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F1-Q1**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

962

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12 cm from A and 6 cm from D

GRADING NOTE Form required: both distances.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

The centroid divides each median so that the part from the vertex is half the part from the midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

963

QUESTION ID

S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The part from the vertex is twice the other part (2 : 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

A. 20

B. 10

C. 7.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

964

QUESTION ID

S23-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

GRADING NOTE BG = 10, so BE = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

965

QUESTION ID

S23-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The centroid is ⅔ of the way from A to the midpoint of BC: A + ⅔((B + C)/2 − A) = (A + B + C)/3.

GRADING NOTE Any equivalent derivation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

967

QUESTION ID

S23-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The three angle bisectors meet at one point I, equidistant from all three sides.

GRADING NOTE REQUIRED WORK: Use the fact that a point on an angle bisector is equidistant from the angle's sides, and its converse.

Any valid proof (paragraph, two-column, or flow). Essential: the bisectors of ∠A and ∠B meet at I; I is equidistant from AB and AC, and from AB and BC; so it is equidistant from AC and BC; by the converse it lies on the bisector of ∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of ∠A and ∠B of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

968

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Each bisector goes from its vertex into the triangle's interior and reaches the opposite side, so the two bisectors cross inside the triangle.

GRADING NOTE Any reasonable explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

A point on the bisector of an angle is equidistant from the two sides of the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

969

QUESTION ID

S23-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved with congruent right triangles (AAS).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

A. circumscribed circle

B. nine-point circle

C. inscribed circle

D. largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

970

QUESTION ID

S23-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. inscribed circle

GRADING NOTE It is equidistant from the sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

The angle bisectors and the medians of every triangle meet at the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

971

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in an equilateral triangle in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

Prove that opposite sides of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Alternate interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 3	Know the definition of a parallelogram	**5.G.B.3**
FORWARD 1	Prove that the diagonals of a rhombus are perpendicular	**HSG-CO.C.11**
FORWARD 2	Use opposite sides with coordinates and algebra	**HSG-GPE.B.4**

SET 24 • HSG-CO.C.11

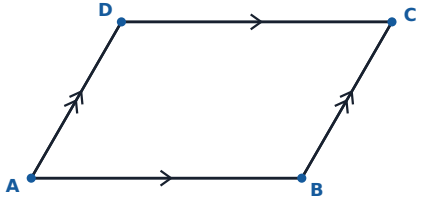
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

974

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB = CD and AD = CB

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △CDA by ASA, using alternate interior angles from both pairs of parallel sides and the shared diagonal.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); ∠BCA ≅ ∠DAC (alternate interior, AD ∥ BC); AC = CA (reflexive); △ABC ≅ △CDA (ASA); corresponding sides. Invalid: "opposite sides look equal"; using the theorem being proved.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram (PQ ∥ SR and PS ∥ QR). Prove that PQ = RS and QR = SP, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

975

QUESTION ID

S24-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PQ = RS and QR = SP

GRADING NOTE REQUIRED WORK: Prove △PQS ≅ △RSQ by ASA with alternate interior angles and the shared diagonal QS.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PQS ≅ ∠RSQ (alternate interior, PQ ∥ SR); ∠PSQ ≅ ∠RQS (alternate interior, PS ∥ QR); QS shared; ASA; corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why AB = CD and AD = CB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

976

QUESTION ID

S24-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

The rotation swaps A and C and sends B to D, so it maps AB onto CD and AD onto CB; rotations keep length.

GRADING NOTE REQUIRED WORK: Find the images of A, line AB, and line AD; then the image of B as an intersection of lines.

Any valid explanation. Essential: M is the midpoint of AC, so A ↔ C; a 180° rotation maps line AB to the parallel line through C, which is line CD (parallel postulate), and line CB to the parallel line through A, which is AD; B = AB ∩ CB goes to CD ∩ AD = D; lengths are kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

977

QUESTION ID

S24-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB = DC = a and AD = BC = √(b² + c²)

GRADING NOTE

REQUIRED WORK: Use the distance formula on each pair of opposite sides.

Any valid proof (paragraph, two-column, or flow). Essential: AB = a; DC = |a + b − b| = a; AD = √(b² + c²); BC = √((a + b − a)² + c²) = √(b² + c²).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

ABCD and ABEF are parallelograms that share side AB, with C, D, E, F all different points. Prove that DC = FE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

978

QUESTION ID

S24-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

DC = AB = FE, so DC = FE.

GRADING NOTE

REQUIRED WORK: Apply the theorem in each parallelogram and use the transitive property.

Any valid proof (paragraph, two-column, or flow). Essential: DC = AB (opposite sides of ABCD); FE = AB (opposite sides of ABEF); transitive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 1

Alternate interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

980

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

∠DCA

GRADING NOTE Any equivalent form. Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

981

QUESTION ID

S24-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Theorem from set 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD with diagonal BD, which angle is congruent to ∠ABD?

A. ∠ABC

B. ∠ADB

C. ∠CBD

D. ∠CDB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

982

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠CDB

GRADING NOTE AB ∥ DC with transversal BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD with diagonal AC, AD ∥ BC. Which angle is congruent to ∠DAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

983

QUESTION ID

S24-B1-Q4

TYPE

Short answer

CORRECT ANSWER

∠BCA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

Alternate interior angles of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

984

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

986

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

ASA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

True or false?

A diagonal shared by two triangles can be the included side in ASA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

987

QUESTION ID

S24-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In △ABC and △CDA, ∠BAC ≅ ∠DCA, ∠BCA ≅ ∠DAC, and AC is shared. Which correspondence is correct?

A. A ↔ D, B ↔ C, C ↔ A

B. A ↔ A, B ↔ D, C ↔ C

C. A ↔ C, B ↔ D, C ↔ A

D. A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

988

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A ↔ C, B ↔ D, C ↔ A

GRADING NOTE Match the equal angles: A ↔ C and C ↔ A, so B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving △ABC ≅ △CDA, which side corresponds to AB?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

989

QUESTION ID
S24-B2-Q4

TYPE
Short answer

CORRECT ANSWER
CD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

True or false?

ASA requires the side to be between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

990

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Otherwise it is AAS.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

True or false?

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

992

QUESTION ID

S24-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

A. A trapezoid with exactly one pair of parallel sides

B. A kite that is not a rhombus

C. Any quadrilateral with one right angle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

993

QUESTION ID

S24-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle

GRADING NOTE A rectangle has both pairs of opposite sides parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

994

QUESTION ID

S24-B3-Q3

TYPE

Short answer

CORRECT ANSWER

PQ and RS; QR and SP

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

True or false?

In parallelogram ABCD, sides AB and BC are opposite sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

995

QUESTION ID

S24-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are consecutive sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Which sides of parallelogram WXYZ are parallel to WX and to XY?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

996

QUESTION ID

S24-B3-Q5

TYPE

Short answer

CORRECT ANSWER

ZY ∥ WX and WZ ∥ XY

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

998

QUESTION ID

S24-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC ⊥ BD

GRADING NOTE REQUIRED WORK: Use that the diagonals of a parallelogram bisect each other, then SSS on triangles AEB and AED (or the perpendicular bisector converse).

Any valid proof (paragraph, two-column, or flow). Essential: BE = ED (diagonals of a parallelogram bisect each other); AB = AD; AE = AE; △AEB ≅ △AED (SSS); ∠AEB ≅ ∠AED form a linear pair, so both are 90°. Alternative: A and C are each equidistant from B and D, so AC is the perpendicular bisector of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

999

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. Half-diagonals 8 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rhombus. Prove that diagonal AC bisects ∠BAD and ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

1000

QUESTION ID

S24-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠BAC ≅ ∠DAC and ∠BCA ≅ ∠DCA

GRADING NOTE REQUIRED WORK: Use SSS on triangles ABC and ADC.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AD and CB = CD (all sides congruent); AC shared; SSS; corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus ABCD has perimeter 52 and diagonal AC = 24. Find diagonal BD.

BD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

1001

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. Side 13, half-diagonals 12 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, m∠PQR = 110°. Find m∠PQS, where QS is a diagonal, and give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

1002

QUESTION ID

S24-F1-Q5

TYPE

Short answer

CORRECT ANSWER

55°, because a diagonal of a rhombus bisects the angles at its ends.

GRADING NOTE Form required: the measure and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 2

Use opposite sides with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(6, 2), and D(3, 5). Find C, and verify that AB = DC and AD = BC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

1004

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

C(8, 5); AB = DC = 5 and AD = BC = √13.

GRADING NOTE Form required: C and both pairs of lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

1005

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Any parallelogram can be moved (rigid motion) so that A is at the origin and AB lies on the positive x-axis; then D = (b, c) and C = B + (D − A) = (a + b, c).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram WXYZ, WX = 3x + 4 and YZ = 5x − 6. Find x and WX.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

1006

QUESTION ID

S24-F2-Q3

TYPE

Short answer

CORRECT ANSWER

x = 5; WX = 19

GRADING NOTE Form required: both. Opposite sides are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

A. √20 + 2

B. 4

C. 2√5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

1007

QUESTION ID

S24-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2√5

GRADING NOTE √(4 + 16) = √20 = 2√5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

1008

QUESTION ID

S24-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(6, 4)

GRADING NOTE Form required: an ordered pair. C = B + (D − A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

SET 101 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that opposite angles of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Same-side interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Name opposite and consecutive angles of a quadrilateral	**5.G.B.3**
BACKWARD 3	Prove triangles congruent with ASA and use corresponding angles	**HSG-CO.B.8**
FORWARD 1	Prove angle properties of special parallelograms	**HSG-CO.C.11**
FORWARD 2	Opposite angles and angles of rotation with coordinates	**HSG-CO.A.3**

SET 101 • HSG-CO.C.11

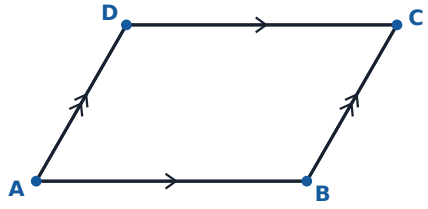
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram.
Prove that ∠B ≅ ∠D.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

1011

QUESTION ID
S101-M-Q1

TYPE
Answer with required work

CORRECT ANSWER
∠B ≅ ∠D

GRADING NOTE REQUIRED WORK: Use diagonal AC and the congruent triangles ABC and CDA (ASA), then corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA and ∠BCA ≅ ∠DAC (alternate interior); AC shared; △ABC ≅ △CDA (ASA); ∠B and ∠D are corresponding angles.

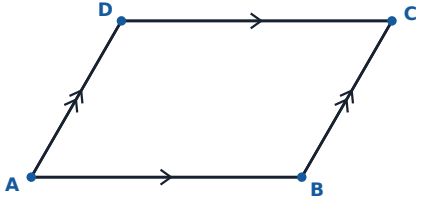
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

1012

QUESTION ID

S101-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠A ≅ ∠C

GRADING NOTE REQUIRED WORK: Show ∠A and ∠B are supplementary and ∠B and ∠C are supplementary, then subtract.

Any valid proof (paragraph, two-column, or flow). Essential: AD ∥ BC with transversal AB gives m∠A + m∠B = 180°; AB ∥ DC with transversal BC gives m∠B + m∠C = 180°; so m∠A = m∠C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram. Prove that ∠P ≅ ∠R, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

1013

QUESTION ID

S101-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

∠P ≅ ∠R

GRADING NOTE REQUIRED WORK: Prove △PQS ≅ △RSQ by ASA, then use corresponding angles.

Any valid proof (paragraph, two-column, or flow). Essential: ∠PQS ≅ ∠RSQ (alternate interior, PQ ∥ SR); ∠PSQ ≅ ∠RQS (alternate interior, PS ∥ QR); QS shared; ASA; ∠P and ∠R correspond.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why ∠DAB ≅ ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

1014

QUESTION ID

S101-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The rotation maps ∠DAB onto ∠BCD, and rotations keep angle measure.

GRADING NOTE REQUIRED WORK: Use that the rotation swaps A and C and maps the sides at A onto the sides at C.

Any valid explanation. Essential: A ↔ C; line AB → the parallel line through C, which is CD; line AD → the parallel line through C, which is CB; ray AB goes to ray CD and ray AD to ray CB, so ∠DAB → ∠BCD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with m∠A = x°. Prove that m∠C = x° and m∠B = m∠D = (180 − x)°, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

1015

QUESTION ID

S101-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

m∠B = 180 − x,
m∠C = x,
m∠D = 180 − x

GRADING NOTE REQUIRED WORK: Use same-side interior angles for each pair of consecutive angles.

Any valid proof (paragraph, two-column, or flow). Essential: m∠A + m∠B = 180° (AD ∥ BC, transversal AB); m∠B + m∠C = 180° (AB ∥ DC); m∠C + m∠D = 180° (AD ∥ BC); solve each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 1

Same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Same-side interior angles formed by a transversal of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

1017

QUESTION ID

S101-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

A. It depends on the figure

B. 180°

C. 360°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

1018

QUESTION ID

S101-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 180°

GRADING NOTE Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

1019

QUESTION ID

S101-B1-Q3

TYPE

Short answer

CORRECT ANSWER

115°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Lines $\ell \parallel m$ are cut by a transversal; one same-side interior angle is 72°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

1020

QUESTION ID
S101-B1-Q4

TYPE
Short answer

CORRECT ANSWER
108°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

1020

True or false?

Same-side interior angles of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

1021

QUESTION ID

S101-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 2

Name opposite and consecutive angles of a quadrilateral • Grade 5 • Geometry • 5.G.B.3

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

1023

QUESTION ID

S101-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠W and ∠Y; ∠X and ∠Z

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrilateral ABCD, ∠A and ∠B are consecutive angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

1024

QUESTION ID

S101-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They share side AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle is opposite ∠Q in quadrilateral PQRS?

A. ∠R

B. ∠P

C. ∠Q

D. ∠S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

1025

QUESTION ID

S101-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. ∠S

GRADING NOTE Opposite vertices are not neighbors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral KLMN: name the angles consecutive to ∠K.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

1026

QUESTION ID

S101-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠L and ∠N

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In quadrilateral ABCD, ∠A and ∠C share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

1027

QUESTION ID

S101-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are opposite angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 3

Prove triangles congruent with ASA and use corresponding angles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC ≅ △CDA with A ↔ C, B ↔ D, C ↔ A. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

1029

QUESTION ID

S101-B3-Q1

TYPE

Short answer

CORRECT ANSWER

∠D

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

True or false?

After two triangles are proved congruent, their corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

1030

QUESTION ID

S101-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of congruence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

Which criterion uses two angles and the side between them?

A. ASA

B. SSS

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

1031

QUESTION ID

S101-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. ASA

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

△PQS ≅ △RSQ with P ↔ R. Which angle corresponds to ∠P?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

1032

QUESTION ID

S101-B3-Q4

TYPE

Short answer

CORRECT ANSWER

∠R

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

True or false?

ASA can use a shared diagonal as its side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

1033

QUESTION ID

S101-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 1

Prove angle properties of special parallelograms • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Prove that a parallelogram with one right angle has four right angles (so it is a rectangle).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

1035

QUESTION ID

S101-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four angles are 90°.

GRADING NOTE REQUIRED WORK: Use opposite angles congruent and consecutive angles supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: the opposite angle is 90° (opposite angles); each consecutive angle is 180° − 90° = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = (3x + 10)° and m∠C = (5x − 30)°. Find x and m∠B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

1036

QUESTION ID

S101-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 20; m∠B = 110°

GRADING NOTE Form required: both. 3x + 10 = 5x − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rhombus ABCD has m∠A = 60°. Prove that diagonal BD divides it into two equilateral triangles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

1037

QUESTION ID

S101-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

△ABD and △CBD each have three 60° angles (and three equal sides).

GRADING NOTE REQUIRED WORK: Use the opposite angle, the equal sides, and the isosceles triangle angle facts.

Any valid proof (paragraph, two-column, or flow). Essential: AB = AD (rhombus), so △ABD is isosceles with vertex angle 60°, and its base angles are (180° − 60°) ÷ 2 = 60°: equilateral; m∠C = 60° (opposite angles) and CB = CD, so △CBD is equilateral too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

In parallelogram PQRS, m∠P is twice m∠Q. Find all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

1038

QUESTION ID

S101-F1-Q4

TYPE

Short answer

CORRECT ANSWER

∠P = ∠R = 120°,
∠Q = ∠S = 60°

GRADING NOTE Form required: all four. 2x + x = 180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

A parallelogram has an angle of 47°. Which list gives all four angles in order?

A. 47°, 43°, 47°, 43°

B. 47°, 47°, 133°, 133°

C. 47°, 133°, 47°, 133°

D. 47°, 133°, 133°, 47°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

1039

QUESTION ID

S101-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 47°, 133°, 47°, 133°

GRADING NOTE Opposite angles are equal; consecutive angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 2

Opposite angles and angles of rotation with coordinates • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). The 180° rotation about (3, 1.5) maps it onto itself. Which angle does ∠A map onto?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

1041

QUESTION ID

S101-F2-Q1

TYPE

Short answer

CORRECT ANSWER

∠C

GRADING NOTE Any equivalent form. A(0, 0) → (6, 3) = C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). Name the image of each vertex under the 180° rotation about (3, 1.5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

1042

QUESTION ID

S101-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A → C, B → D, C → A, D → B

GRADING NOTE Form required: all four. (x, y) → (6 − x, 3 − y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

True or false?

A 180° rotation about the center of a parallelogram maps each angle onto the angle opposite it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

1043

QUESTION ID

S101-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It swaps opposite vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

Which transformation shows that opposite angles of every parallelogram are congruent?

A. A 90° rotation about a vertex

B. A translation along a side

C. A reflection across a diagonal

D. A 180° rotation about the intersection of the diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

1044

QUESTION ID

S101-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A 180° rotation about the intersection of the diagonals

GRADING NOTE Reflections across diagonals work only for rhombuses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram PQRS has P(1, 1), Q(5, 1), R(7, 4), S(3, 4). Give the center and angle of a rotation that maps ∠P onto ∠R.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

1045

QUESTION ID

S101-F2-Q5

TYPE

Short answer

CORRECT ANSWER

180° about (4, 2.5)

GRADING NOTE Form required: the center and the angle. The center is the midpoint of PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that the diagonals of a parallelogram bisect each other

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Opposite sides of a parallelogram are congruent	**HSG-CO.C.11**
BACKWARD 2	Vertical and alternate interior angles at the crossing of the diago…	**HSG-CO.C.9**
BACKWARD 3	Find the midpoint of a segment and use the definition of bisect	**6.G.A.3**
FORWARD 1	Prove that a 180° rotation carries a parallelogram onto itself	**HSG-CO.A.3**
FORWARD 2	Use diagonal bisection with coordinates and algebra	**HSG-GPE.B.4**

SET 25 • HSG-CO.C.11

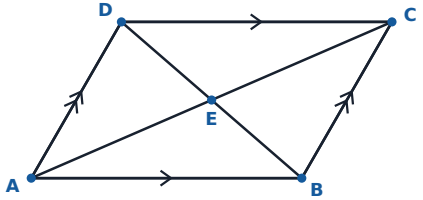
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

1048

QUESTION ID

S25-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Prove △ABE ≅ △CDE by ASA (or AAS), using alternate interior angles and AB = CD.

Any valid proof (paragraph, two-column, or flow). Essential: AB = CD (opposite sides of a parallelogram); ∠BAE ≅ ∠DCE and ∠ABE ≅ ∠CDE (alternate interior, AB ∥ DC); △ABE ≅ △CDE (ASA); corresponding sides. Vertical angles at E may replace one angle pair (AAS). Invalid: assuming E is a midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

1049

QUESTION ID

S25-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

PT = TR and QT = TS, so T is the midpoint of both diagonals.

GRADING NOTE REQUIRED WORK: Prove △PTQ ≅ △RTS, using alternate interior angles and PQ = RS.

Any valid proof (paragraph, two-column, or flow). Essential: PQ = RS; two pairs of alternate interior angles (PQ ∥ SR); ASA; corresponding sides; definition of midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

1050

QUESTION ID

S25-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

Both diagonals have midpoint ((a + b)/2, c/2).

GRADING NOTE

REQUIRED WORK: Show the two diagonals have the same midpoint.

Any valid proof (paragraph, two-column, or flow). Essential: midpoint of AC = ((a + b)/2, c/2); midpoint of BD = ((a + b)/2, c/2); the same point lies on both diagonals and is the midpoint of each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with diagonals meeting at E. Prove that AE = CE and BE = DE using triangles AED and CEB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

1051

QUESTION ID

S25-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AE = CE and BE = DE

GRADING NOTE REQUIRED WORK: Use AD = CB, alternate interior angles from AD ∥ BC, and ASA.

Any valid proof (paragraph, two-column, or flow). Essential: AD = CB (opposite sides); ∠DAE ≅ ∠BCE and ∠ADE ≅ ∠CBE (alternate interior, AD ∥ BC); △AED ≅ △CEB (ASA); corresponding sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

WXYZ is a parallelogram whose diagonals WY and XZ meet at V. Prove that V is the midpoint of WY and of XZ, using the 180° rotation about the midpoint M of WY (show that M is also the midpoint of XZ, so M = V).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

1052

QUESTION ID

S25-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

The rotation swaps W and Y and maps X to Z, so M is the midpoint of XZ; the diagonals meet at M, so V = M.

GRADING NOTE REQUIRED WORK: Show that the rotation maps X to Z; then M is the midpoint of XZ too.

Any valid explanation. Essential: W ↔ Y; line WX → the parallel line through Y, which is YZ; line YX → the parallel line through W, which is WZ; X = WX ∩ YX → YZ ∩ WZ = Z; so M is the midpoint of XZ, and M lies on both diagonals, so it is their intersection V.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

1054

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

CD = 9, DA = 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

True or false?

Opposite sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

1055

QUESTION ID

S25-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved in the previous set.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

1056

QUESTION ID

S25-B1-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

A. PQ

B. PR

C. SP

D. RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

1057

QUESTION ID

S25-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. SP

GRADING NOTE Opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

True or false?

Consecutive sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

1058

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

1060

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠CED

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

1061

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

∠DCA

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

Vertical angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

1062

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Proved earlier.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

A. ∠CDB

B. ∠ABC

C. ∠ADB

D. ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

1063

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. ∠CDB

GRADING NOTE Alternate interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

∠AEB and ∠BEC are vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

1064

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They form a linear pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

1066

QUESTION ID

S25-B3-Q1

TYPE

Short answer

CORRECT ANSWER

((a + b)/2, c/2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

1067

QUESTION ID

S25-B3-Q2

TYPE

Short answer

CORRECT ANSWER

((a + b)/2, c/2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

A segment bisects another segment if it passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

1068

QUESTION ID

S25-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of bisect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

1069

QUESTION ID

S25-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

True or false?

If two segments have the same midpoint, each bisects the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

1070

QUESTION ID

S25-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Their common point is the midpoint of both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

1072

QUESTION ID

S25-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

A ↔ C and B ↔ D, so the parallelogram maps onto itself.

GRADING NOTE REQUIRED WORK: Use the fact that E is the midpoint of both diagonals to find the images of the vertices.

Any valid proof (paragraph, two-column, or flow). Essential: a 180° rotation about E maps each point P to the point P′ with E the midpoint of PP′; E is the midpoint of AC and BD, so A → C, C → A, B → D, D → B; segments map to segments, so each side maps onto its opposite side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

1073

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 5)

GRADING NOTE Form required: an ordered pair. E is the midpoint of AC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has diagonals that bisect each other at E. Explain why the 180° rotation about E carries ABCD onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

1074

QUESTION ID

S25-F1-Q3

TYPE

Short answer

CORRECT ANSWER

E is the midpoint of AC and of BD, so the rotation swaps A with C and B with D, and it maps each side onto the opposite side.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

A. side SP

B. side RS

C. side QR

D. diagonal PR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

1075

QUESTION ID

S25-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. side RS

GRADING NOTE P → R and Q → S.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

A 180° rotation about the intersection of its diagonals carries a parallelogram onto itself. Explain why this implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

1076

QUESTION ID

S25-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The rotation maps each angle onto the opposite angle, and rotations keep angle measures.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

1078

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

D(3, 6); diagonals meet at (5, 4)

GRADING NOTE Form required: both points. Midpoint of AC = (5, 4) = midpoint of BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

1079

QUESTION ID
S25-F2-Q2

TYPE
Short answer

CORRECT ANSWER
26

GRADING NOTE Any equivalent form. 3x − 2 = x + 8 gives x = 5; AE = 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

1080

QUESTION ID

S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They bisect each other but are congruent only in a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

A. (4, 3)

B. (3, 3)

C. (5, 3)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

1081

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (5, 3)

GRADING NOTE (0, 0), (4, 0), (5, 3), (1, 3) is a parallelogram (others are possible: (3, −3), (−3, 3)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

1082

QUESTION ID
S25-F2-Q5

TYPE
Short answer

CORRECT ANSWER
30

GRADING NOTE Any equivalent form. y = 5, QT = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove the converses: conditions that make a quadrilateral a parallelogram

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Converses of the parallel-line angle theorems	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with SSS and SAS	**HSG-CO.B.8**
BACKWARD 3	Angle sum of a quadrilateral	**8.G.A.5**
FORWARD 1	Prove that a quadrilateral is a parallelogram using coordinates	**HSG-GPE.B.4**
FORWARD 2	Decide which conditions are enough to guarantee a parallelogram	**HSG-CO.C.11**

SET 26 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

1085

QUESTION ID

S26-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △CDA by SSS, then use the converse of the alternate interior angles theorem twice.

Any valid proof (paragraph, two-column, or flow). Essential: SSS (AB = CD, BC = DA, AC shared); ∠BAC ≅ ∠DCA, so AB ∥ DC (converse); ∠BCA ≅ ∠DAC, so BC ∥ AD; definition. Invalid: using properties of parallelograms before proving it is one.

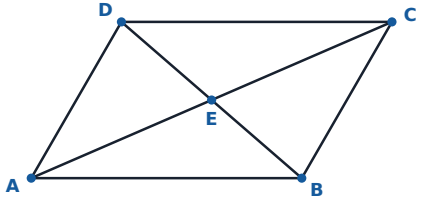
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

1086

QUESTION ID

S26-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Prove △AEB ≅ △CED and △AED ≅ △CEB by SAS (vertical angles), then use congruent alternate interior angles.

Any valid proof (paragraph, two-column, or flow). Essential: AE = CE, BE = DE (bisect); vertical angles; SAS twice; ∠EAB ≅ ∠ECD gives AB ∥ DC; ∠EAD ≅ ∠ECB gives AD ∥ BC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

1087

QUESTION ID

S26-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC, so with AB ∥ DC, ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Draw diagonal AC; prove △ABC ≅ △CDA by SAS, then show AD ∥ BC.

Any valid proof (paragraph, two-column, or flow). Essential: ∠BAC ≅ ∠DCA (alternate interior, AB ∥ DC); AB = CD; AC shared; SAS; ∠BCA ≅ ∠DAC; AD ∥ BC (converse).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, ∠A ≅ ∠C and ∠B ≅ ∠D. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

1088

QUESTION ID

S26-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC and AB ∥ DC.

GRADING NOTE REQUIRED WORK: Use the quadrilateral angle sum to show consecutive angles are supplementary, then the converse of the same-side interior angles theorem.

Any valid proof (paragraph, two-column, or flow). Essential: 2m∠A + 2m∠B = 360°, so m∠A + m∠B = 180°; same-side interior angles supplementary gives AD ∥ BC; similarly m∠A + m∠D = 180° gives AB ∥ DC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In quadrilateral ABCD, ∠A and ∠B are supplementary, and ∠B and ∠C are supplementary. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

1089

QUESTION ID

S26-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

AD ∥ BC and AB ∥ DC, so ABCD is a parallelogram.

GRADING NOTE REQUIRED WORK: Use the converse of the same-side interior angles theorem twice.

Any valid proof (paragraph, two-column, or flow). Essential: with transversal AB, same-side interior angles ∠A and ∠B are supplementary, so AD ∥ BC; with transversal BC, ∠B and ∠C are supplementary, so AB ∥ DC; definition of a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

1091

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The lines are parallel.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

1092

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

A. Congruent corresponding angles

B. A linear pair

C. Two acute angles

D. Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

1093

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Congruent corresponding angles

GRADING NOTE Corresponding angles converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

1094

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes (they add to 180°).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If alternate interior angles are 40° and 50°, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

1095

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They must be congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

△ABC and △CDA share AC, and AB = CD, BC = DA.
Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

1097

QUESTION ID
S26-B2-Q1

TYPE
Short answer

CORRECT ANSWER
SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

1098

QUESTION ID
S26-B2-Q2

TYPE
Short answer

CORRECT ANSWER
SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

True or false?

Vertical angles can serve as the included angle in SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

1099

QUESTION ID

S26-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After △ABC ≅ △CDA, which angle is congruent to ∠BAC?

A. ∠BCA

B. ∠ABC

C. ∠DAC

D. ∠DCA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

1100

QUESTION ID

S26-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ∠DCA

GRADING NOTE A ↔ C, B ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

True or false?

SSA proves two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

1101

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not a valid criterion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

1103

QUESTION ID

S26-B3-Q1

TYPE

Short answer

CORRECT ANSWER

75°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

True or false?

The angles of every quadrilateral add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

1104

QUESTION ID

S26-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A diagonal splits it into two triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

1105

QUESTION ID

S26-B3-Q3

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE Any equivalent form. 6x = 360.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

A. 90°

B. 70°

C. 110°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

1106

QUESTION ID

S26-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 110°

GRADING NOTE 360 − 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

True or false?

A quadrilateral can have four acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

1107

QUESTION ID

S26-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Four angles under 90° add to less than 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

1109

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

It is a parallelogram: the diagonals AC and BD have the same midpoint (4.5, 3.5), so they bisect each other.

GRADING NOTE Any valid method (slopes of opposite sides, or opposite side lengths, or diagonal midpoints) with the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

1110

QUESTION ID
S26-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Not a parallelogram: PQ has slope 1/4 but SR has slope 1/5, so PQ is not parallel to SR.

GRADING NOTE Any valid evidence and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

True or false?

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

1111

QUESTION ID
S26-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

A. AC and BD have equal lengths

B. AB ⊥ BC

C. AB and DC have equal slopes and equal lengths

D. AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

1112

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. AB and DC have equal slopes and equal lengths

GRADING NOTE One pair of opposite sides parallel and congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

1113

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. DC must equal AB = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q5**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

1115

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. An isosceles trapezoid is a counterexample.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

1116

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The angle sum makes consecutive angles supplementary, so both pairs of opposite sides are parallel.

GRADING NOTE Any valid explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

1117

QUESTION ID

S26-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A kite has one diagonal bisecting the other but is not a parallelogram in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

A. One pair of opposite sides congruent

B. Both pairs of opposite sides congruent

C. Diagonals bisect each other

D. One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

1118

QUESTION ID

S26-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One pair of opposite sides congruent

GRADING NOTE One pair of congruent sides alone allows many shapes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

1119

QUESTION ID

S26-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No. A kite (not a rhombus) has perpendicular diagonals but is not a parallelogram.

GRADING NOTE Any valid counterexample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that rectangles are parallelograms with congruent diagonals

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Parallelogram properties: opposite sides and consecutive angles	**HSG-CO.C.11**
BACKWARD 2	SAS and SSS with overlapping triangles	**HSG-CO.B.8**
BACKWARD 3	Use the Pythagorean Theorem for diagonals	**8.G.B.7**
FORWARD 1	Decide whether four points form a rectangle	**HSG-GPE.B.4**
FORWARD 2	Prove diagonal properties of rhombuses and squares	**HSG-CO.C.11**

SET 27 • HSG-CO.C.11

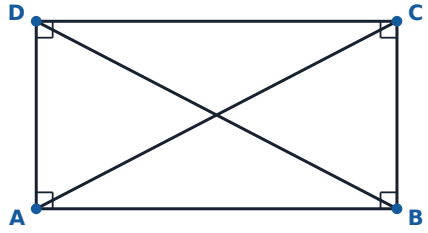
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a rectangle. Prove that AC = BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

1122

QUESTION ID

S27-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC = BD

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △BAD by SAS (right angles, shared side AB, and AD = BC).

Any valid proof (paragraph, two-column, or flow). Essential: a rectangle is a parallelogram, so AD = BC; ∠ABC ≅ ∠BAD (right angles); AB shared; SAS; corresponding sides AC = BD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

1123

QUESTION ID

S27-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

All four angles are right angles, so ABCD is a rectangle.

GRADING NOTE REQUIRED WORK: Prove △ABC ≅ △BAD by SSS, show ∠A ≅ ∠B, then use that consecutive angles are supplementary.

Any valid proof (paragraph, two-column, or flow). Essential: BC = AD (opposite sides); AC = BD (given); AB shared; SSS; ∠DAB ≅ ∠CBA; they are supplementary (consecutive angles), so each is 90°; opposite angles then are 90° too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

1124

QUESTION ID

S27-M-Q3

TYPE

Answer with required work

CORRECT ANSWER

$AC = BD = \sqrt{(a^2 + b^2)}$

GRADING NOTE REQUIRED WORK: Use the distance formula on both diagonals.

Any valid proof (paragraph, two-column, or flow). Essential: $AC = \sqrt{(a^2 + b^2)}$ and $BD = \sqrt{(a^2 + b^2)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Quadrilateral ABCD has four right angles. Prove that ABCD is a parallelogram (so a rectangle is a parallelogram).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

1125

QUESTION ID

S27-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

AB ∥ DC and AD ∥ BC.

GRADING NOTE REQUIRED WORK: Use the converse of the same-side interior angles theorem.

Any valid proof (paragraph, two-column, or flow). Essential: ∠A + ∠B = 180° with transversal AB, so AD ∥ BC; ∠A + ∠D = 180° with transversal AD, so AB ∥ DC (or: both are perpendicular to the same line). Invalid: "it looks like a parallelogram".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The diagonals of quadrilateral ABCD bisect each other and are congruent. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

1126

QUESTION ID

S27-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a parallelogram with congruent diagonals, so it is a rectangle.

GRADING NOTE REQUIRED WORK: First show ABCD is a parallelogram (diagonals bisect each other), then use the parallelogram-with-congruent-diagonals theorem or SSS.

Any valid proof (paragraph, two-column, or flow). Essential: diagonals that bisect each other make ABCD a parallelogram (set 26); then $\triangle ABC \cong \triangle BAD$ by SSS (BC = AD, AC = BD, AB shared), so $\angle A \cong \angle B$; consecutive angles are supplementary, so each is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

1128

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°, 90°, 90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

1129

QUESTION ID

S27-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same-side interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

1130

QUESTION ID

S27-B1-Q3

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form. They are also supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

A. AC

B. CD

C. AB

D. BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

1131

QUESTION ID

S27-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. BC

GRADING NOTE Opposite sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

True or false?

In every parallelogram, all angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

1132

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only in rectangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

1134

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

∠BAD

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

True or false?

Overlapping triangles can share a side used in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

1135

QUESTION ID

S27-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Reflexive property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

A. SSS

B. ASA

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

1136

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. SSS

GRADING NOTE Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

1137

QUESTION ID

S27-B2-Q4

TYPE

Short answer

CORRECT ANSWER

SAS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

True or false?

In △ABC ≅ △BAD, vertex C corresponds to vertex C.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

1138

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C ↔ D.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

1140

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

1141

QUESTION ID

S27-B3-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

True or false?

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

1142

QUESTION ID

S27-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

A. 8

B. 4√2

C. 2√2

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

1143

QUESTION ID

S27-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4√2

GRADING NOTE √32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

True or false?

The two diagonals of a 3-by-7 rectangle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

1144

QUESTION ID

S27-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both are √58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

1146

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Rectangle: it is a parallelogram (AB ∥ DC, slope ½; AD ∥ BC, slope −2) and AB ⊥ AD (½ · −2 = −1).

GRADING NOTE Any valid method (or: diagonals bisect each other and are congruent, both 5 units long).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

1147

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Not a rectangle: it is a parallelogram, but its diagonals PR = √34 and QS = √18 are not congruent.

GRADING NOTE Any valid evidence and the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

1148

QUESTION ID
S27-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Parallelogram with congruent diagonals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

A. Diagonals bisect each other and are congruent

B. Diagonals are congruent

C. One angle is a right angle

D. Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

1149

QUESTION ID

S27-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Diagonals bisect each other and are congruent

GRADING NOTE Congruent diagonals alone fit an isosceles trapezoid.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

1150

QUESTION ID

S27-F1-Q5

TYPE

Short answer

CORRECT ANSWER

C(6, 4);
AC = BD = √52
= 2√13.

GRADING NOTE Form required: C and both diagonal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

1152

QUESTION ID

S27-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

All four sides are congruent.

GRADING NOTE REQUIRED WORK: Use that the diagonals bisect each other, then SAS on two adjacent triangles.

Any valid proof (paragraph, two-column, or flow). Essential: diagonals bisect at E; right angles at E; shared segment; SAS gives AB = AD; opposite sides give the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

1153

QUESTION ID
S27-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Its diagonals bisect each other, are congruent, and are perpendicular.

GRADING NOTE Any equivalent complete list.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

True or false?

A rectangle with congruent diagonals is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

1154

QUESTION ID

S27-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Every rectangle has congruent diagonals; a square needs perpendicular diagonals too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

A. rectangle that is not a square

B. rhombus that is not a square

C. square

D. trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

1155

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. square

GRADING NOTE Rectangle (congruent) and rhombus (perpendicular).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

1156

QUESTION ID

S27-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Side $5\sqrt{2}$; area 50

GRADING NOTE Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

Construct a copy of a segment

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Compare and measure lengths	**2.MD.A.1**
BACKWARD 2	A compass draws all points at one distance from its center	**7.G.B.4**
BACKWARD 3	Know rays, segments, and their endpoints	**HSG-CO.A.1**
FORWARD 1	Construct a triangle from three given segments	**HSG-CO.D.12**
FORWARD 2	Decide whether segments can form a triangle, and construct sums	**HSG-CO.D.12**

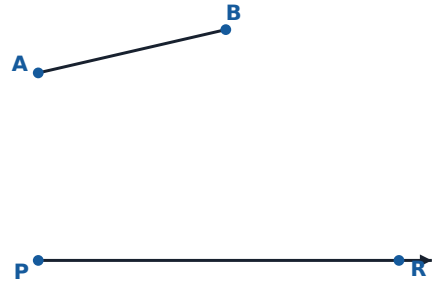
SET 28 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

1159

QUESTION ID

S28-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A compass arc centered at P with radius AB crosses ray PR at Q; PQ = AB.

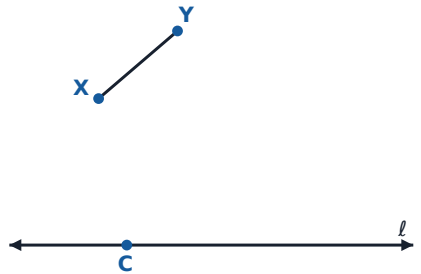
GRADING NOTE Grade the construction marks, not only the final figure: the compass set to AB (an arc or tick at A and B is allowed) and an arc centered at P meeting the ray at Q, which is labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line ℓ, starting at point C, construct a segment CD with CD = XY.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

1160

QUESTION ID

S28-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An arc centered at C with radius XY crosses ℓ at D (either side of C is correct).

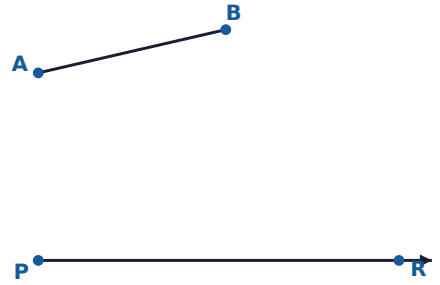
GRADING NOTE Grade the construction marks, not only the final figure: compass opening equal to XY, arc centered at C, D marked on ℓ. Either intersection point is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

1161

QUESTION ID

S28-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Copy AB from P to get M on the ray, then copy AB again from M (away from P) to get Q; PQ = 2AB.

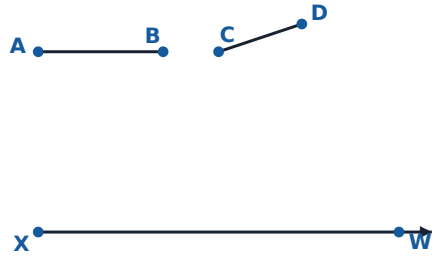
GRADING NOTE Grade the construction marks, not only the final figure: two arcs of radius AB, one centered at P and one centered at the first point, with Q labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

On ray XW, construct a point Z so that XZ = AB + CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

1162

QUESTION ID

S28-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy AB from X to get Y on the ray, then copy CD from Y (away from X) to get Z; XZ = AB + CD.

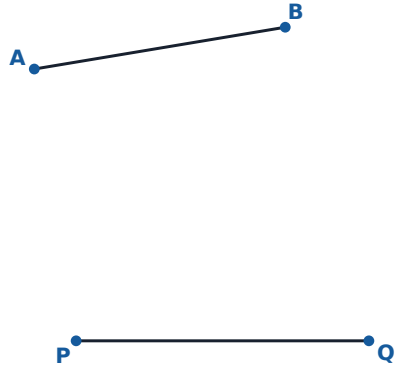
GRADING NOTE Grade the construction marks, not only the final figure: an arc of radius AB centered at X, an arc of radius CD centered at the first point, and Z labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct an isosceles triangle with base PQ whose two legs are each congruent to segment AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

1163

QUESTION ID

S28-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs of radius AB centered at P and at Q meet at R; triangle PQR has PR = QR = AB.

GRADING NOTE Grade the construction marks, not only the final figure: two arcs with the compass set to AB, centered at P and Q, their crossing R, and the legs drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

1165

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer: equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Two segments are congruent if they have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

1166

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of congruent segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

A. A compass

B. A calculator

C. A protractor

D. A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

1167

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A compass

GRADING NOTE The compass keeps a fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

1168

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A segment drawn longer on the page than another is congruent to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

1169

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent segments have equal lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

1171

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

True or false?

Changing the compass opening between steps of a construction keeps the copy accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

1172

QUESTION ID

S28-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The opening must stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

An arc centered at P crosses a ray from P at Q. What is PQ?

A. It cannot be known

B. Half the opening

C. Twice the opening

D. The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

1173

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The compass opening

GRADING NOTE Q is on the arc, so it is one radius from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

Every point on a circle is the same distance from the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

1174

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

1175

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Ray PR starts at P and continues through R without end.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

1177

QUESTION ID

S28-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition of a ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

Q is on ray PR with PQ = 5. Must Q be between P and R? Describe all the cases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

1178

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

No. If PR > 5, Q is between P and R; if PR = 5, Q is R; if PR < 5, R is between P and Q.

GRADING NOTE Form required: all three cases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

A. none

B. only B

C. only A

D. A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

1179

QUESTION ID

S28-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A and B

GRADING NOTE A segment has two endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

True or false?

A point on ray PR can be on the opposite side of P from R.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

1180

QUESTION ID
S28-B3-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The ray goes only from P toward R.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

1181

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

P

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

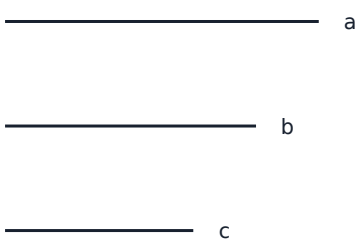
SET 28 • HSG-CO.D.12

FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

1183

QUESTION ID

S28-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a as base XY; arcs centered at X with radius b and at Y with radius c meet at Z; triangle XYZ has sides a, b, c.

GRADING NOTE Grade the construction marks, not only the final figure: the copied base, two arcs with radii b and c, and their intersection joined to the base endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

1184

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The third vertex must be b from one end of the base and c from the other, so it lies on both circles, where the arcs cross.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

True or false?

Any three segments can be used to construct a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

1185

QUESTION ID

S28-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The two arcs meet only if each length is less than the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

To construct a triangle from three given segments, arcs are drawn from the two ends of the base, and they cross at two points, one on each side of the base. The two triangles formed are

A. similar but not congruent

B. different in size

C. not triangles

D. congruent (mirror images)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

1186

QUESTION ID

S28-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. congruent (mirror images)

GRADING NOTE Reflection across the base swaps them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

1187

QUESTION ID

S28-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Yes. 3 + 4 > 6, so the arcs meet.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

1189

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. 2 + 3 < 6, so the arcs of radii 2 and 3 centered at the ends of the 6 base do not meet.

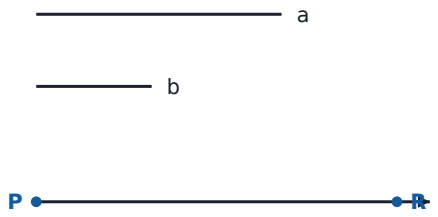
GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

1190

QUESTION ID

S28-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a from P to Q on the ray; copy b from Q back toward P to S; PS = a − b.

GRADING NOTE Grade the construction marks, not only the final figure: two compass arcs (radius a at P, radius b at Q toward P), with the result labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

True or false?

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

1191

QUESTION ID

S28-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then one length equals the sum of the other two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

A. 1, 2, 3

B. 3, 4, 5

C. 4, 4, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

1192

QUESTION ID

S28-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1, 2, 3

GRADING NOTE 1 + 2 = 3, so the figure is flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

1193

QUESTION ID
S28-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Any length between 3 and 13 (3 < x < 13).

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 102 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Make formal constructions with a variety of tools and methods

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Congruent segments have equal lengths	**2.MD.A.1**
BACKWARD 2	A reflection maps a point to its mirror image	**8.G.A.1**
FORWARD 1	Choose and justify a method for a construction	**HSG-CO.D.12**
FORWARD 2	Locate the circumcenter by folding	**HSG-C.A.3**

SET 102 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line ℓ starting at point C, and explain why the copy is exact.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

1196

QUESTION ID

S102-M-Q1

TYPE

Short answer

CORRECT ANSWER

Lay the strip along AB and mark the positions of A and B on its edge. Move the strip to ℓ, put the A-mark on C along ℓ, and mark the point at the B-mark: that is D. The strip carries the same length without measuring, so CD = AB.

GRADING NOTE Any valid method using a strip (or string) to transfer the length without reading a ruler, with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

Describe how to construct the perpendicular bisector of segment AB by folding paper, and explain why the crease is the perpendicular bisector.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

1197

QUESTION ID

S102-M-Q2

TYPE

Short answer

CORRECT ANSWER

Fold so that A lands exactly on B and crease. The fold is a reflection that maps A to B, so by the definition of a reflection the crease is the perpendicular bisector of AB.

GRADING NOTE Any equivalent method and reason (the fold matches A onto B; a reflection line is the perpendicular bisector of PP′).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

Describe how to bisect an angle drawn on paper by folding, and explain why the crease bisects the angle.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

1198

QUESTION ID

S102-M-Q3

TYPE

Short answer

CORRECT ANSWER

Fold through the vertex so that one side lies exactly on the other side. The fold is a reflection that maps one side onto the other and fixes the vertex, so it makes two congruent angles.

GRADING NOTE Any equivalent method and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

You have a pencil, a pin, and a piece of string, but no compass. Describe how to construct an equilateral triangle on segment AB, and explain why it works.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

1199

QUESTION ID

S102-M-Q4

TYPE

Short answer

CORRECT ANSWER

Tie the string to the pin so the length from pin to pencil is AB. Pin it at A and draw an arc; pin it at B and draw an arc; the arcs cross at C.
AC = BC = AB because the string length did not change.

GRADING NOTE Any valid string method that keeps one fixed length, with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In dynamic geometry software, describe how to construct the line through P parallel to line ℓ so that it stays parallel when P or ℓ is dragged, and explain why a line drawn "by eye" is not a construction.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

1200

QUESTION ID

S102-M-Q5

TYPE

Short answer

CORRECT ANSWER

Use the parallel-line tool (or construct a perpendicular to ℓ through P, then a perpendicular to that line at P). The software keeps the defining relationship when points move; a line placed by eye is not tied to ℓ and stops being parallel when anything is dragged.

GRADING NOTE Any valid software construction based on a defining relationship, and the drag-test reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 1

Congruent segments have equal lengths • Grade 2 • Measurement & Data • 2.MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Segment PQ is 4.5 cm and segment RS is 45 mm. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

1202

QUESTION ID

S102-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 45 mm = 4.5 cm, so the lengths are equal.

GRADING NOTE Any equivalent answer with the unit conversion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A strip of paper can carry a length from one place to another without a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

1203

QUESTION ID

S102-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Mark the two ends on the strip.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

Which of these can carry a length from one segment to another without a ruler?

A. A thermometer

B. A calculator

C. A folded paper strip marked at both ends

D. A protractor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

1204

QUESTION ID

S102-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A folded paper strip marked at both ends

GRADING NOTE The marks keep a fixed length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

A string is cut to the length of segment AB. It is laid along line ℓ from C. How long is the segment it marks?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

1205

QUESTION ID

S102-B1-Q4

TYPE

Short answer

CORRECT ANSWER

AB (the same length)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

True or false?

Two segments with different lengths can be congruent if they are drawn with the same tool.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

1206

QUESTION ID

S102-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Congruent segments have equal lengths, whatever tool is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 2

A reflection maps a point to its mirror image • Grade 8 • Geometry • 8.G.A.1

True or false?

Folding paper along a line matches each point with its reflection across the fold.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

1208

QUESTION ID

S102-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The fold line is the mirror.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

A paper is folded so that point A lands on point B. What is the fold line, in relation to segment AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

1209

QUESTION ID

S102-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflection keeps

A. neither

B. lengths and angle measures

C. only lengths

D. only angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

1210

QUESTION ID

S102-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. lengths and angle measures

GRADING NOTE Rigid motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

True or false?

A point on the fold line moves when the paper is folded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

1211

QUESTION ID

S102-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on the fold line stay fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Folding makes ray BA land on ray BC. What is true of ∠ABD and ∠DBC, where BD is the crease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

1212

QUESTION ID

S102-B2-Q5

TYPE

Short answer

CORRECT ANSWER

They are congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 1

Choose and justify a method for a construction • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which tool or method would you use to find the midpoint of a segment drawn on tracing paper, without a compass? Explain why it is exact.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

1214

QUESTION ID

S102-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Fold A onto B: the crease is the perpendicular bisector, so it crosses AB at the midpoint.

GRADING NOTE Any valid method with the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

A reflective device (a transparent mirror) is placed on line ℓ, and point P is seen in it. Explain how to use it to locate the reflection of P across ℓ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

1215

QUESTION ID

S102-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Look through the device placed along ℓ and mark the point where the image of P appears behind it; that point is P′, the reflection of P.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why measuring a 60° angle with a protractor is not a formal construction, but stepping a compass around a circle is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

1216

GRADING NOTE Any equivalent explanation.

QUESTION ID

S102-F1-Q3

TYPE

Short answer

CORRECT ANSWER

A formal construction uses only the tools' exact operations (fixed lengths, lines through points), not readings of a scale; the protractor reading is an approximation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe a way to construct the perpendicular from P to line ℓ by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

1217

QUESTION ID

S102-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Fold the paper so that ℓ lands on itself and the crease passes through P; the crease is perpendicular to ℓ.

GRADING NOTE Any valid folding method with ℓ folded onto itself through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In dynamic geometry software, a student constructs a "square" by placing four points by eye. Describe a drag test that shows it is not a constructed square, and how to construct one properly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

1218

QUESTION ID

S102-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Drag a vertex: the figure stops being a square. Construct it from a segment with perpendiculars at both ends and circles of radius equal to the side, so it stays a square when dragged.

GRADING NOTE Any valid test and construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 2

Locate the circumcenter by folding • High School — Geometry • Geometry: Circles • HSG-C.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Describe how to find the point equidistant from the three vertices of a paper triangle by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

1220

QUESTION ID

S102-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Fold each of two vertices onto another (A onto B, then B onto C); the two creases are perpendicular bisectors and cross at the circumcenter.

GRADING NOTE Any valid folding method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

Why do two creases suffice, without folding the third pair of vertices?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

1221

QUESTION ID

S102-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The crossing point is equidistant from A and B and from B and C, so it is equidistant from A and C and lies on the third bisector too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular-bisector creases of a triangle cross can lie outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

1222

QUESTION ID

S102-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For an obtuse triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

Folding vertex A onto vertex B of a triangle gives a crease that is

A. the perpendicular bisector of AB

B. the bisector of ∠C

C. the median from C

D. parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

1223

QUESTION ID

S102-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the perpendicular bisector of AB

GRADING NOTE The fold reflects A onto B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

After finding the circumcenter O by folding, how can you draw the circumscribed circle with a pin and string?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

1224

QUESTION ID

S102-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Pin the string at O and set its length to OA; draw the circle through A, which also passes through B and C.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a copy of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Copy a segment with a compass	**HSG-CO.D.12**
BACKWARD 3	Arcs and circles in constructions	**7.G.B.4**
FORWARD 1	Justify why the angle-copy construction works	**HSG-CO.B.8**
FORWARD 2	Use angle copying to construct a parallel line	**HSG-CO.D.12**

SET 29 • HSG-CO.D.12

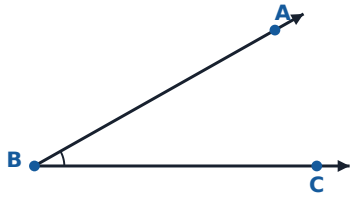
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.





REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

1227

QUESTION ID

S29-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at B meets the sides at D and E; same arc centered at P meets ray PR at F; arc centered at F with radius DE meets the first arc at G; ray PG.

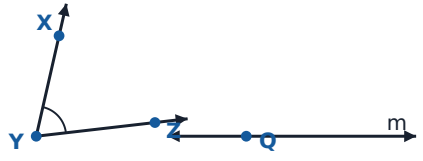
GRADING NOTE Grade the construction marks, not only the final figure: the arc at B, the same-radius arc at P, the arc with radius DE centered on ray PR, and ray PG drawn. The copy may open above or below ray PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

1228

QUESTION ID

S29-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps as copying an angle: arcs of the same radius at Y and Q, then the chord length transferred; the new ray from Q opens upward.

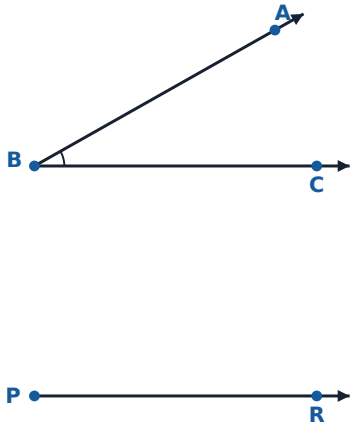
GRADING NOTE Grade the construction marks, not only the final figure: equal-radius arcs at Y and Q and the transferred chord arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC, opening BELOW ray PR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

1229

QUESTION ID

S29-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Equal-radius arcs at B and at P, the chord DE transferred from F on the lower side, and ray PG drawn below ray PR.

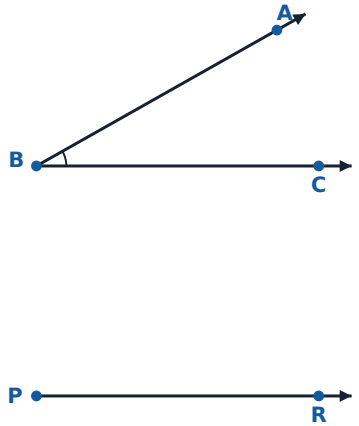
GRADING NOTE Grade the construction marks, not only the final figure: the arc at B, the same-radius arc at P, the chord arc centered on ray PR, and the new ray below PR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

1230

QUESTION ID

S29-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy ∠ABC onto ray PR to get ray PG, then copy ∠ABC again onto ray PG (on the other side from R) to get ray PH; ∠RPH = 2∠ABC.

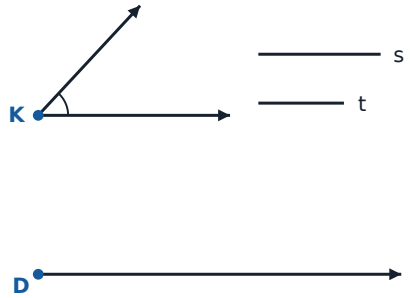
GRADING NOTE Grade the construction marks, not only the final figure: two complete angle copies sharing ray PG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct triangle DEF with ∠D ≅ ∠K, DE = s, and DF = t.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

1231

QUESTION ID

S29-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Copy ∠K at D with one side on the ray; copy s along the ray from D to get E; copy t along the other side from D to get F; join E and F.

GRADING NOTE Grade the construction marks, not only the final figure: the complete angle copy at D and the two segment copies; triangle DEF drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

1233

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

True or false?

An angle's measure does not change when its sides are drawn longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

1234

QUESTION ID

S29-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Measure depends only on the opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which angle is obtuse?

A. 90°

B. 45°

C. 130°

D. 20°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

1235

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 130°

GRADING NOTE Between 90° and 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

1236

QUESTION ID

S29-B1-Q4

TYPE

Short answer

CORRECT ANSWER

70°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

True or false?

Two angles with sides of different lengths cannot be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

1237

QUESTION ID

S29-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only the measures matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

To copy ∠ABC to vertex P, an arc centered at B crosses BA at D and BC at E, and the same arc centered at P crosses ray PR at F. Which length is set on the compass next, to locate the second side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

1239

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

DE, the chord between the two arc crossings

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

True or false?

To copy a length, set the compass to the length and do not change it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

1240

QUESTION ID

S29-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fixed opening.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

A. Half of DE

B. BD

C. DE

D. PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

1241

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. DE

GRADING NOTE G is on the circle about F with radius DE.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

1242

QUESTION ID

S29-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

True or false?

A segment copy can be checked only by measuring with a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

1243

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Compass marks are the evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

True or false?

Two arcs centered at different points can cross at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

1245

QUESTION ID

S29-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Two circles can meet twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

1246

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

BD = BE = 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

A. Part of a circle

B. A ray

C. A straight segment

D. A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

1247

QUESTION ID

S29-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Part of a circle

GRADING NOTE The compass draws circles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

True or false?

An arc centered at P passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

1248

QUESTION ID

S29-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Points on it are one radius away from P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

1249

QUESTION ID

S29-B3-Q5

TYPE

Short answer

CORRECT ANSWER

The center of the arc

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

1251

QUESTION ID

S29-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠DBE ≅ ∠FPG

GRADING NOTE REQUIRED WORK: Prove △DBE ≅ △FPG by SSS, then use corresponding angles.

Any valid proof. Essential: BD = PF, BE = PG, DE = FG (construction); SSS; corresponding angles ∠DBE and ∠FPG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

∠XYZ is copied to vertex Q, opening downward: an arc of radius r centered at Y meets the sides at U and V, the same arc centered at Q meets the given ray at U′, and an arc of radius UV centered at U′ meets it at V′. Prove that ∠U′QV′ ≅ ∠XYZ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

1252

QUESTION ID

S29-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

∠U′QV′ ≅ ∠UYV, which is ∠XYZ.

GRADING NOTE

REQUIRED WORK: Name the three pairs of equal segments, then use SSS.

Any valid proof. Essential: YU = QU′ = r, YV = QV′ = r, UV = U′V′ (construction); △UYV ≅ △U′QV′ (SSS); corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

∠ABC is copied twice at P: first onto ray PR to give ray PG, then onto ray PG (on the side away from R) to give ray PH. Prove that m∠RPH = 2 · m∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

1253

QUESTION ID

S29-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

m∠RPH = m∠RPG + m∠GPH = 2m∠ABC

GRADING NOTE REQUIRED WORK: Use the justified copy twice and the angle addition postulate.

Any valid proof. Essential: each copy is congruent to ∠ABC (SSS justification); G is in the interior of ∠RPH, so the measures add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

In copying ∠ABC to P, a student keeps the arc radius but sets the compass to BD (a radius) instead of the chord DE for the last arc. Which step of the SSS justification fails, and what goes wrong with the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

1254

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

The third pair of sides is no longer DE = FG, so SSS does not apply; the new ray makes a 60° angle (an equilateral triangle) instead of ∠ABC unless ∠ABC is 60°.

GRADING NOTE Any equivalent explanation that names the missing side equality.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Explain why the angle-copy construction works using a rigid motion instead of SSS: there is a rigid motion taking B to P and ray BC onto ray PR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

1255

QUESTION ID

S29-F1-Q5

TYPE

Answer with required work

CORRECT ANSWER

The motion sends E to F (same radius on ray PR) and D to the point on the arc at P that is DE from F on the chosen side, which is G; so ∠ABC → ∠FPG.

GRADING NOTE REQUIRED WORK: Follow D, E, F, G under the rigid motion and use the fact that two circles meet in at most one point on each side of a line.

Any valid explanation. Essential: rigid motions keep the distances BD, BE, DE; the image of D is on the arc at P and DE from F; on one side of line PR that point is G.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

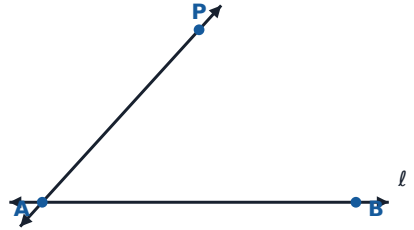
SET 29 • HSG-CO.D.12

FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

1257

QUESTION ID

S29-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Copy the angle that line AP makes with ℓ at A to vertex P, in the corresponding position; the new side through P is parallel to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: a complete angle copy at P (equal-radius arcs and chord) in the corresponding position, and the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

1258

QUESTION ID
S29-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Congruent corresponding angles make the lines parallel (converse of the corresponding angles theorem).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

True or false?

Copying the angle so the two angles are congruent same-side interior angles also produces a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

1259

QUESTION ID

S29-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Same-side interior angles must be supplementary, not congruent (unless both are 90°).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

To construct the line through P parallel to ℓ by copying an angle, a transversal is drawn through P meeting ℓ. Which angle is copied?

A. The angle between ℓ and itself

B. The angle the transversal makes with ℓ

C. A right angle

D. Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

1260

QUESTION ID
S29-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The angle the transversal makes with ℓ

GRADING NOTE The transversal through P and a point of ℓ creates the angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

1261

QUESTION ID

S29-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Construct the perpendicular from P to ℓ, then construct the perpendicular to that line at P; it is parallel to ℓ.

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the perpendicular bisector of a segment (bisect a segment)

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Points equidistant from A and B lie on the perpendicular bisector	**HSG-CO.C.9**
BACKWARD 2	Know midpoint and perpendicular bisector definitions	**HSG-CO.A.1**
BACKWARD 3	Arcs with one radius from two centers	**7.G.B.4**
FORWARD 1	Justify the perpendicular bisector construction	**HSG-CO.C.9**
FORWARD 2	Construct the circumcenter and circumscribed circle of a triangle	**HSG-C.A.3**

SET 30 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

1264

QUESTION ID

S30-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs of equal radius (more than half of AB) centered at A and at B meet at two points; the line through them is the perpendicular bisector, meeting AB at M.

GRADING NOTE Grade the construction marks, not only the final figure: two equal-radius arcs from A and B crossing on both sides of AB, the line through the crossings, and M labeled.

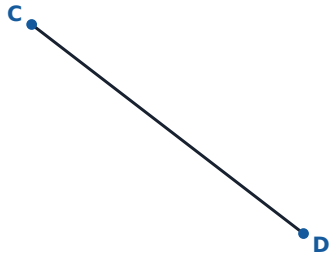
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

1265

QUESTION ID

S30-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The perpendicular bisector construction locates M where the line through the two arc crossings meets CD.

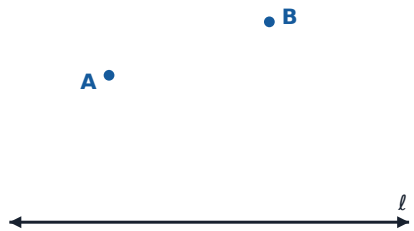
GRADING NOTE Grade the construction marks, not only the final figure: the two arc crossings and the line through them; M labeled on CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line ℓ that is the same distance from A as from B.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

1266

QUESTION ID

S30-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisector of AB; the point P where it meets ℓ is equidistant from A and B.

GRADING NOTE Grade the construction marks, not only the final figure: the full perpendicular bisector construction for AB and its intersection with ℓ marked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of the short vertical segment UV.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

1267

QUESTION ID

S30-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Equal arcs (radius more than half of UV) centered at U and V cross on the left and right; the horizontal line through the crossings bisects UV.

GRADING NOTE Grade the construction marks, not only the final figure: two equal-radius arcs from each endpoint, the two crossings, and the line through them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Divide segment AB into four congruent parts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

1268

QUESTION ID

S30-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect AB at M; then bisect AM and MB; the three points cut AB into four equal parts.

GRADING NOTE Grade the construction marks, not only the final figure: three perpendicular bisector constructions (for AB, AM, and MB).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

1270

QUESTION ID

S30-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse of the perpendicular bisector theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

1271

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the perpendicular bisector construction for segment AB, the two arc crossings are equidistant from A and B because

A. the crossings are on AB

B. the arcs have the same radius

C. the compass is opened to AB

D. the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

1272

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. the arcs have the same radius

GRADING NOTE Each crossing is one radius from A and one radius from B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

1273

QUESTION ID
S30-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two points determine more than one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

1274

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one line passes through two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

1276

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector of a segment passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

1277

QUESTION ID

S30-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

A. the perpendicular bisector

B. not a bisector

C. parallel to AB

D. a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

1278

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. a bisector but not the perpendicular bisector

GRADING NOTE It bisects but is not perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

1279

QUESTION ID

S30-B2-Q4

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A segment has many perpendicular bisectors in one plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

1280

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

1282

QUESTION ID

S30-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Yes, at two points (5 + 5 > 6).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

True or false?

If the compass opening is less than half of AB, the arcs from A and B do not meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

1283

QUESTION ID

S30-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The two radii add to less than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

Why must the opening be more than half of AB?

A. It does not matter

B. So the midpoint is on the arc

C. So the arcs from A and B cross

D. So the line is longer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

1284

QUESTION ID

S30-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. So the arcs from A and B cross

GRADING NOTE Equal radii must add to more than AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

1285

QUESTION ID

S30-B3-Q4

TYPE

Short answer

CORRECT ANSWER

XA = XB = r

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

True or false?

The two arc crossings are on the same side of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

1286

QUESTION ID

S30-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One is on each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

To construct the perpendicular bisector of AB, arcs of radius r (more than half of AB) centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

1288

QUESTION ID

S30-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

PQ is the perpendicular bisector of AB.

GRADING NOTE REQUIRED WORK: Use the converse of the perpendicular bisector theorem for P and for Q.

Any valid proof. Essential: PA = PB = r and QA = QB = r; each of P and Q is on the perpendicular bisector (converse); two points determine that line. Alternative: SSS gives congruent triangles APQ and BPQ, then SAS/linear pair at the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

Explain why the perpendicular bisector construction for AB would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

1289

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Any equal radius greater than half of AB gives two points equidistant from A and B.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

True or false?

The construction works only if the arcs are drawn with radius exactly AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

1290

QUESTION ID

S30-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any equal radius over half of AB works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

A. Midsegment theorem

B. Triangle angle sum

C. Converse of the perpendicular bisector theorem

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

1291

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Converse of the perpendicular bisector theorem

GRADING NOTE Equidistant points lie on the bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

1292

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The fold is a reflection swapping A and B, so the crease is the perpendicular bisector of AB (definition of reflection).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

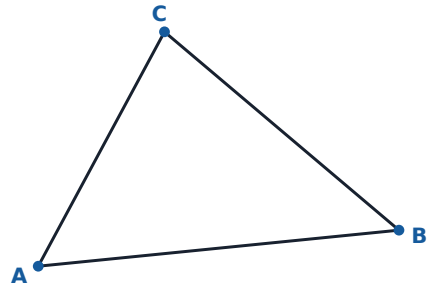
SET 30 • HSG-CO.D.12

FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

1294

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular bisectors of two sides; they meet at O; the circle centered at O through A also passes through B and C.

GRADING NOTE Grade the construction marks, not only the final figure: two complete perpendicular bisector constructions, O labeled, and the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

1295

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Their crossing is equidistant from all three vertices, so it is on the third bisector too.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

True or false?

The circumcenter of an obtuse triangle lies on its longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

1296

QUESTION ID

S30-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It lies outside, beyond the longest side; it is on the longest side only for a right triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

A. Inside, near the centroid

B. At the midpoint of the hypotenuse

C. Outside the triangle

D. At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

1297

QUESTION ID

S30-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. At the midpoint of the hypotenuse

GRADING NOTE The hypotenuse is a diameter of the circumcircle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

1298

QUESTION ID

S30-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(4, 3); radius 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the bisector of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of an angle bisector	**HSG-CO.A.1**
BACKWARD 2	Prove triangles congruent with SSS	**HSG-CO.B.8**
BACKWARD 3	Measure angles with a protractor	**4.MD.C.6**
FORWARD 1	Justify the angle bisector construction	**HSG-CO.B.8**
FORWARD 2	Construct special angles and the incenter by bisecting	**HSG-CO.D.12**

SET 31 • HSG-CO.D.12

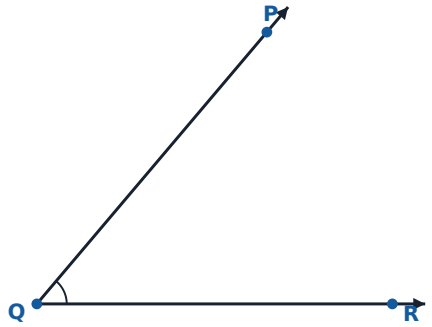
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the bisector of ∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

1301

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at Q crosses the sides at X and Y; equal arcs centered at X and Y meet at Z inside the angle; ray QZ is the bisector.

GRADING NOTE Grade the construction marks, not only the final figure: the arc at Q, two equal-radius arcs from X and Y crossing inside the angle, and ray QZ.

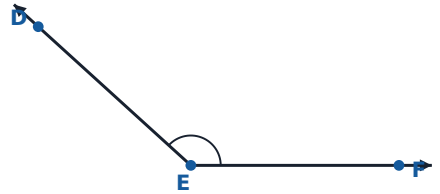
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

1302

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same steps: arc at E, equal arcs from the two crossing points, ray from E through their intersection.

GRADING NOTE Grade the construction marks, not only the final figure: all three arcs and the bisector ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

1303

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular to ℓ at A (bisect the straight angle), then bisect one of the right angles.

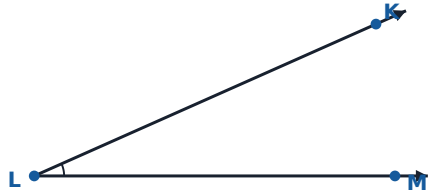
GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction at A and the angle bisector construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the bisector of the narrow angle ∠KLM.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

1304

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Arc centered at L crosses both sides; equal arcs from those points cross inside the angle; ray from L through the crossing.

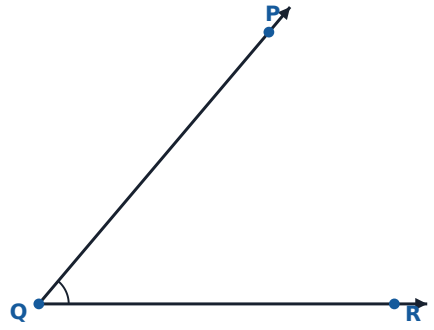
GRADING NOTE Grade the construction marks, not only the final figure: the arc at L, the two equal arcs, and the bisector ray.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a ray QS inside ∠PQR so that m∠RQS = ¼ m∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

1305

QUESTION ID

S31-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect ∠PQR with ray QT, then bisect ∠TQR with ray QS.

GRADING NOTE Grade the construction marks, not only the final figure: two complete angle bisector constructions, the second inside the half next to ray QR.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

1307

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

True or false?

An angle bisector divides the angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

1308

QUESTION ID

S31-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects ∠ABC with m∠ABC = 3x + 6 and m∠ABD = 24. Find x.

A. 6

B. 10

C. 18

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

1309

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 14

GRADING NOTE 3x + 6 = 48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

1310

QUESTION ID

S31-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

True or false?

An angle bisector is a segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

1311

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a ray from the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

1313

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

SSS

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

True or false?

SSS guarantees that corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

1314

QUESTION ID

S31-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Congruent triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

To bisect ∠ABC, an arc centered at B crosses BA at X and BC at Y, and equal arcs centered at X and Y meet at Z. Which triangles are congruent?

A. △ABC and △XYZ

B. △BXZ and △BYZ

C. △BXY and △XYZ

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

1315

QUESTION ID
S31-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. △BXZ and △BYZ

GRADING NOTE BX = BY, XZ = YZ, BZ shared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

In the angle-bisector construction for ∠ABC (arc at B meeting the sides at X and Y, equal arcs from X and Y meeting at Z), △BXZ ≅ △BYZ. Which angle is congruent to ∠XBZ?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

1316

QUESTION ID

S31-B2-Q4

TYPE

Short answer

CORRECT ANSWER

∠YBZ

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

True or false?

SSS requires knowing one angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

1317

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only three sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

1319

QUESTION ID

S31-B3-Q1

TYPE

Short answer

CORRECT ANSWER

56°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

True or false?

Measuring with a protractor is a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

1320

QUESTION ID

S31-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Constructions use compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

An angle of 150° is bisected. Each part is

A. 60°

B. 150°

C. 75°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

1321

QUESTION ID

S31-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75°

GRADING NOTE 150 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

1322

QUESTION ID

S31-B3-Q4

TYPE

Short answer

CORRECT ANSWER

76°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

True or false?

A protractor can check a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

1323

QUESTION ID

S31-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Measuring can check, though it does not replace the construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

ANSWER KEY Reduced copy of the question page, with its answer

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

1325

QUESTION ID

S31-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

∠XBZ ≅ ∠YBZ

GRADING NOTE REQUIRED WORK: Prove △BXZ ≅ △BYZ by SSS.

Any valid proof. Essential: BX = BY (same arc), XZ = YZ (equal arcs), BZ shared; SSS; corresponding angles; definition of bisector.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

In the angle bisector construction, an arc centered at the vertex crosses the sides at X and Y, and then arcs from X and Y cross at Z. Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

1326

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

So that XZ = YZ, which SSS needs.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

True or false?

The bisector construction works for obtuse angles too.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

1327

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The SSS argument does not depend on the size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

A. AAS

B. ASA

C. SSS

D. SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

1328

QUESTION ID

S31-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. SSS

GRADING NOTE Three pairs of sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

1329

QUESTION ID

S31-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The fold is a reflection that maps one side onto the other, so it makes two congruent angles at the vertex.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

1331

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct an equilateral triangle on AB (60° angle at A), then bisect the angle at A to get 30°.

GRADING NOTE Grade the construction marks, not only the final figure: arcs of radius AB from A and B, the 60° angle, and its bisector.

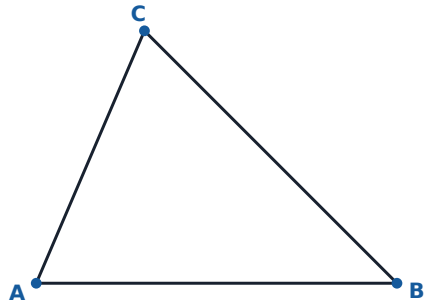
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

1332

QUESTION ID

S31-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Bisect two angles of the triangle; their intersection is the incenter.

GRADING NOTE Grade the construction marks, not only the final figure: two complete angle bisector constructions and the crossing labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

True or false?

Bisecting an angle and then bisecting each half divides the angle into three equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

1333

QUESTION ID

S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives four equal parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

A. 15°

B. 75°

C. 20°

D. 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

1334

QUESTION ID

S31-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 20°

GRADING NOTE Bisecting gives 30°, 15°, 45°, 22.5°, ... and 75° = 60° + 15°. 20° would require trisecting 60°, which these steps cannot do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

1335

QUESTION ID

S31-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Construct a right angle, bisect it to get 45°, and add the 45° angle to the right angle (90° + 45°).

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a perpendicular to a line through a point on or off the line

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of perpendicular lines	**HSG-CO.A.1**
BACKWARD 2	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 3	Recognize perpendicular lines in figures	**4.G.A.1**
FORWARD 1	Construct an altitude of a triangle and use it for area	**HSG-CO.D.12**
FORWARD 2	Construct a square on a given segment	**HSG-CO.D.12**

SET 32 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is on ℓ.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

1338

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs centered at P mark A and B on ℓ equally far from P; then construct the perpendicular bisector of AB, which passes through P.

GRADING NOTE Grade the construction marks, not only the final figure: the two equal marks on ℓ and the perpendicular bisector arcs, with the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is not on ℓ.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

1339

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

An arc centered at P crosses ℓ at A and B; equal arcs from A and B meet at Q below ℓ; line PQ is perpendicular to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: the arc from P meeting ℓ twice, equal arcs from A and B, and line PQ.

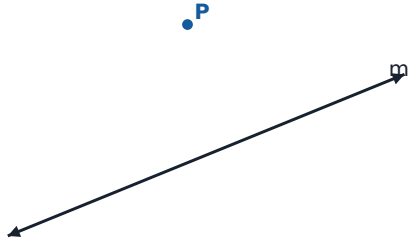
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

1340

QUESTION ID

S32-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Same method as for a point not on the line: arc from P meets m twice, equal arcs from those points meet, and the line through P and the crossing.

GRADING NOTE Grade the construction marks, not only the final figure: all arcs and the perpendicular line.

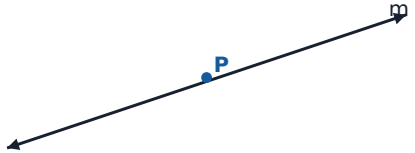
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

1340

Compass and straightedge only; leave all arcs.

Construct the line perpendicular to the slanted line m at the point P on m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

1341

QUESTION ID

S32-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Arcs centered at P mark two points on m equally far from P; the perpendicular bisector of those two points passes through P.

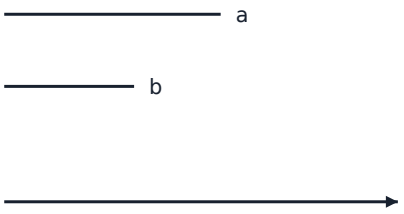
GRADING NOTE Grade the construction marks, not only the final figure: the two equal marks on m and the perpendicular bisector arcs, with the line drawn through P.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a right triangle whose legs are congruent to segments a and b.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

1342

QUESTION ID

S32-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Copy a onto the ray as XY; construct the perpendicular to the ray at X; copy b onto it as XZ; join Y and Z.

GRADING NOTE Grade the construction marks, not only the final figure: the copy of a, the perpendicular construction at X, the copy of b, and the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Perpendicular lines meet at a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

1344

QUESTION ID

S32-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

Line k ⊥ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

1345

QUESTION ID
S32-B1-Q2

TYPE
Short answer

CORRECT ANSWER
90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of lines is perpendicular?

A. y = 3 and y = −3

B. y = 2x and y = 2x + 1

C. y = x and y = 2x

D. y = 2x and y = −½x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

1346

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x and y = −½x

GRADING NOTE Slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Through a point, there are many lines perpendicular to a given line in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

1347

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S32-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q4/5

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

1348

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The perpendicular from the point to the line

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on ℓ, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

1350

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of AB

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

1351

QUESTION ID

S32-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Equidistant crossings.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

A. equidistant from A and B

B. on segment AB

C. on line ℓ

D. at A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

1352

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. equidistant from A and B

GRADING NOTE Same radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P is on ℓ, and A and B are marked on ℓ with PA = PB.
Where does the perpendicular bisector of AB cross ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

1353

QUESTION ID
S32-B2-Q4

TYPE
Short answer

CORRECT ANSWER
At P

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of AB passes through A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

1354

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It passes through the midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

True or false?

A square's adjacent sides are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

1356

QUESTION ID

S32-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

A. V

B. T

C. O

D. S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

1357

QUESTION ID

S32-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. T

GRADING NOTE T has a right angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

1358

QUESTION ID

S32-B3-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

True or false?

Two lines that look almost perpendicular must be perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

1359

QUESTION ID

S32-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Appearance is not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

1360

QUESTION ID

S32-B3-Q5

TYPE

Short answer

CORRECT ANSWER

A vertical side and the top (or bottom) edge

GRADING NOTE Any correct pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

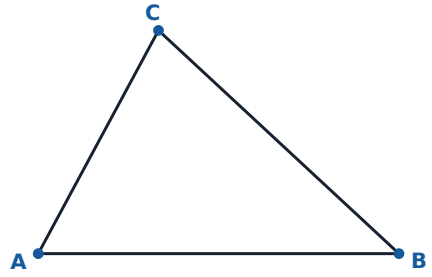
SET 32 • HSG-CO.D.12

FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

1362

QUESTION ID

S32-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular from C to line AB; the segment from C to the foot is the altitude.

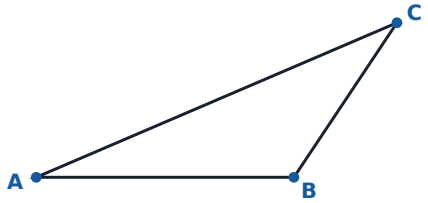
GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction from C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

1363

QUESTION ID

S32-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Extend line AB past B and construct the perpendicular from C to that line.

GRADING NOTE Grade the construction marks, not only the final figure: AB extended and the perpendicular construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

1364

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

30

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

True or false?

Every altitude of a triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

1365

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An obtuse triangle has two altitudes outside it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

A. the hypotenuse

B. inside the triangle

C. outside the triangle

D. a leg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

1366

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. inside the triangle

GRADING NOTE It meets the hypotenuse between its endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a square with side AB, above AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

1368

QUESTION ID

S32-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct perpendiculars to AB at A and at B; copy AB onto each, on the SAME side of AB (above it), to get D and C; join C and D.

GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular constructions and two segment copies on the same side of AB. Copies on opposite sides give a crossed figure, not a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A square on segment AB is constructed by drawing perpendiculars to AB at A and at B and copying AB onto each, on the same side of line AB, to get D (from A) and C (from B). Explain why ABCD is a square, and why the two copies must be on the same side of AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

1369

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

AD ∥ BC (both ⊥ AB) and AD = BC, so ABCD is a parallelogram; it has a right angle at A, so it is a rectangle; AB = AD, so all sides are equal: a square. If D and C were on opposite sides, segment DC would cross AB and ABCD would not be a quadrilateral with these sides.

GRADING NOTE Any valid explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with four congruent sides is always a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

1370

QUESTION ID

S32-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A rhombus need not have right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

A. Bisect AB

B. Construct a 60° angle

C. Draw a circle through A and B

D. Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

1371

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Copy AB twice along the base

GRADING NOTE The base must be doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

1372

QUESTION ID

S32-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$5\sqrt{2}$

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a line parallel to a given line through a point not on it

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Copy an angle with compass and straightedge	**HSG-CO.D.12**
BACKWARD 2	Converse: congruent corresponding angles give parallel lines	**HSG-CO.C.9**
BACKWARD 3	Recognize parallel lines	**4.G.A.1**
FORWARD 1	Justify the parallel-line constructions	**HSG-CO.C.9**
FORWARD 2	Divide a segment into equal parts with parallel lines	**HSG-SRT.B.4**

SET 33 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

1375

QUESTION ID

S33-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a transversal through P meeting ℓ at A; copy the angle it makes with ℓ at A to P in the corresponding position; the new side is parallel to ℓ.

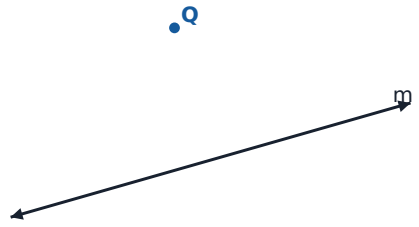
GRADING NOTE Grade the construction marks, not only the final figure: a transversal, a complete angle copy at P (equal-radius arcs and transferred chord), and the parallel line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

1376

QUESTION ID

S33-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Same method: a transversal through Q, then copy the corresponding (or alternate interior) angle at Q.

GRADING NOTE Grade the construction marks, not only the final figure: all arcs of the angle copy and the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to ℓ using two perpendiculars.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

1377

QUESTION ID

S33-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular k from P to ℓ, then the perpendicular to k at P; it is parallel to ℓ.

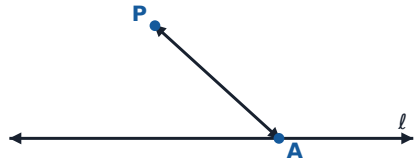
GRADING NOTE Grade the construction marks, not only the final figure: two complete perpendicular constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an ALTERNATE INTERIOR angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

1378

QUESTION ID

S33-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Copy the angle between ℓ and line AP at A to vertex P, on the opposite side of line AP (alternate interior position); the new side through P is parallel to ℓ.

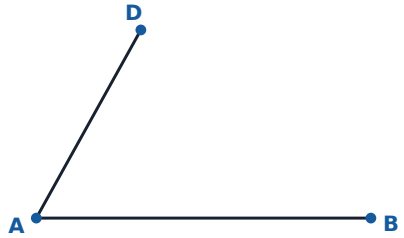
GRADING NOTE Grade the construction marks, not only the final figure: a complete angle copy at P in the alternate interior position and the line drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct parallelogram ABCD with sides AB and AD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

1379

QUESTION ID

S33-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the line through D parallel to AB and the line through B parallel to AD; they meet at C.

GRADING NOTE Grade the construction marks, not only the final figure: two parallel-line constructions (angle copies or two perpendiculars) and the point C labeled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

ANSWER KEY Reduced copy of the question page, with its answer

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

1381

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The radius of the first arc (twice) and the chord between the side crossings

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

True or false?

An angle copy uses arcs of the same radius at both vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

1382

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

A. Draw a circle

B. Measure with a protractor

C. Draw the ray from the new vertex through the last arc crossing

D. Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

1383

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Draw the ray from the new vertex through the last arc crossing

GRADING NOTE The ray is the new side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

1384

QUESTION ID

S33-B1-Q4

TYPE

Short answer

CORRECT ANSWER

SSS makes the two triangles congruent.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

True or false?

An angle copy needs a protractor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

1385

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles mean the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

1387

QUESTION ID

S33-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

1388

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

A. Congruent vertical angles

B. Supplementary linear pairs

C. Congruent alternate interior angles

D. Two acute angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

1389

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Congruent alternate interior angles

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Same-side interior angles of 70° and 70° prove the lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

1390

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They must add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

1391

QUESTION ID

S33-B2-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

The opposite edges of a ruler are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

1393

QUESTION ID

S33-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE They never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

A. V

B. A

C. H

D. X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

1394

QUESTION ID

S33-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. H

GRADING NOTE H's two uprights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

1395

QUESTION ID

S33-B3-Q3

TYPE

Short answer

CORRECT ANSWER

Parallel lines

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Lines that cross are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

1396

QUESTION ID

S33-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Parallel lines never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In a plane, how many lines through a point NOT on a given line are parallel to that line?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

1397

QUESTION ID

S33-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Parallel postulate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

To construct a line through P parallel to ℓ, a transversal through P meets ℓ at A, and the angle at A is copied at P in the corresponding position. Prove that the new line is parallel to ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

1399

QUESTION ID

S33-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The new line $\parallel \ell$.

GRADING NOTE REQUIRED WORK: Use the congruence of the copied angle (SSS) and the converse of the corresponding angles theorem.

Any valid proof. Essential: the copy is congruent (SSS from the construction); the angles are corresponding angles along AP; the converse gives parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the two-perpendicular construction, k ⊥ ℓ and n ⊥ k at P. Prove that n ∥ ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

1400

QUESTION ID

S33-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

n ∥ ℓ

GRADING NOTE REQUIRED WORK: Both n and ℓ make right angles with k (corresponding angles congruent), or two lines perpendicular to the same line in a plane are parallel.

Any valid proof (paragraph, two-column, or flow). Essential: n ⊥ k and ℓ ⊥ k give a pair of congruent (90°) corresponding angles where k crosses n and ℓ; congruent corresponding angles give n ∥ ℓ. Alternative: two lines perpendicular to the same line are parallel (with that theorem cited). Invalid: "they look parallel".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

True or false?

Copying the alternate interior angle at P also gives a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

1401

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Alternate interior converse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to ℓ?

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

B. Bisect the angle at P

C. Draw a circle centered at P

D. Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

1402

QUESTION ID

S33-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

GRADING NOTE Only the angle copy gives congruent corresponding angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

1403

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Exactly one (the unique parallel through P).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

1405

QUESTION ID

S33-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a ray from A; mark three equal steps C, D, E on it with the compass; join E to B; construct parallels to EB through D and C; they cut AB into three equal parts.

GRADING NOTE Grade the construction marks, not only the final figure: three equal compass steps on the auxiliary ray and two parallel constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

To divide AB into three equal parts, three equal steps are marked on a ray from A, the last mark is joined to B, and parallels to that segment are drawn through the other marks. Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

1406

QUESTION ID
S33-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Parallel lines cut the two sides of the triangle proportionally (triangle proportionality), so equal steps on the ray give equal parts of AB.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

True or false?

The same method can divide a segment into 5 equal parts only if its length is a multiple of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

1407

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal compass steps on the auxiliary ray work for any segment; nothing is measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

A. 2

B. 5

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

1408

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE 2 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

1409

QUESTION ID

S33-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct an equilateral triangle inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Central angles of a regular polygon	**HSG-CO.A.3**
BACKWARD 2	Copy a length with a compass	**HSG-CO.D.12**
BACKWARD 3	Circle vocabulary: center, radius, chord, diameter	**7.G.B.4**
FORWARD 1	Justify that the inscribed triangle is equilateral	**HSG-C.A.2**
FORWARD 2	Construct a regular dodecagon by bisecting arcs	**HSG-CO.D.13**

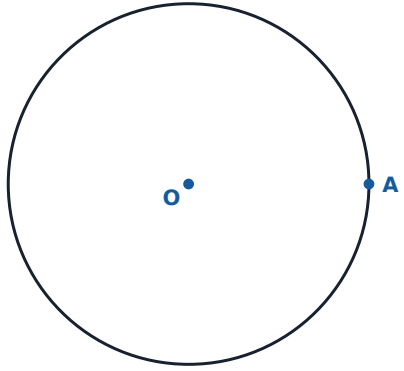
SET 34 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

1412

QUESTION ID

S34-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

With the compass set to the radius, step around the circle from A marking six points; join every other point (A, the 2nd, the 4th).

GRADING NOTE Grade the construction marks, not only the final figure: radius-length arcs stepped around the circle (or the diameter-and-arcs method) and the triangle joining alternate points.

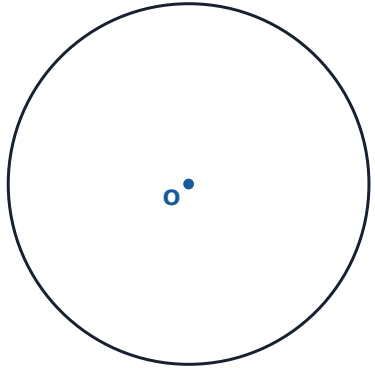
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

1413

QUESTION ID

S34-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Mark a point on the circle, step the radius around the circle to get six points, and join alternate points.

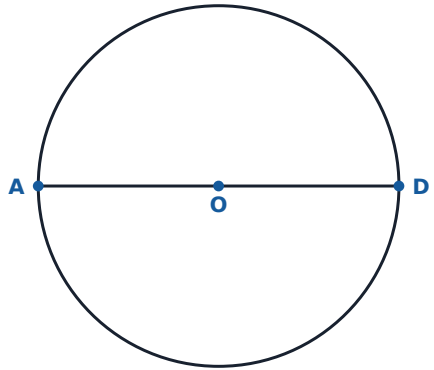
GRADING NOTE Grade the construction marks, not only the final figure: stepping arcs and the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

1414

QUESTION ID

S34-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The arc centered at D with radius DO meets the circle at B and C; triangle ABC is equilateral.

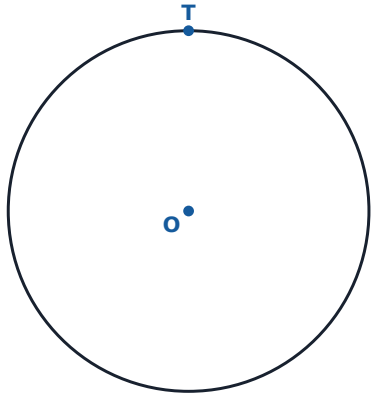
GRADING NOTE Grade the construction marks, not only the final figure: the arc at D and the triangle ABC.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle with one vertex at the top point T.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

1415

QUESTION ID

S34-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Step the radius around the circle from T to mark six points; join T to the 2nd and 4th points.

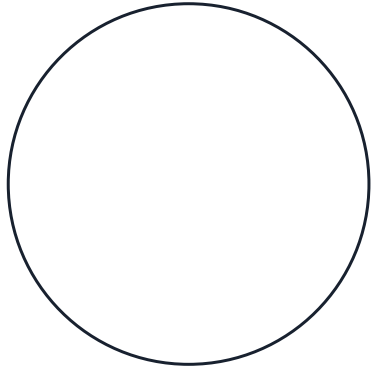
GRADING NOTE Grade the construction marks, not only the final figure: stepping arcs from T and the triangle joining alternate points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct an equilateral triangle inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

1416

QUESTION ID

S34-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Draw two chords and construct their perpendicular bisectors; they meet at the center O; then step the radius around the circle and join alternate points.

GRADING NOTE Grade the construction marks, not only the final figure: two perpendicular bisector constructions locating O, the compass set to the radius, and the triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

1418

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

120°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

1419

QUESTION ID
S34-B1-Q2

TYPE
Short answer

CORRECT ANSWER
6 (each 60°)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The vertices of an inscribed equilateral triangle are 120° apart around the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

1420

QUESTION ID

S34-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 360 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

A. a square

B. a rectangle

C. an equilateral triangle

D. a pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

1421

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. an equilateral triangle

GRADING NOTE Every other vertex: 120° apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The central angles of an inscribed equilateral triangle are 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

1422

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

1424

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A chord equal to the radius

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

True or false?

Keeping the compass opening fixed is required when stepping around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

1425

QUESTION ID

S34-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE All chords must be equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

A. a square

B. an equilateral triangle

C. a right triangle

D. an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

1426

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. an equilateral triangle

GRADING NOTE All three sides equal the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

1427

QUESTION ID

S34-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

True or false?

Stepping a compass set to the diameter marks six points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

1428

QUESTION ID
S34-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The diameter fits only twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

1430

QUESTION ID

S34-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A segment with both endpoints on the circle

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

True or false?

A diameter is a chord through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

1431

QUESTION ID

S34-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

A circle has radius 6. Its diameter is

A. 3

B. 36

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

1432

QUESTION ID

S34-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

GRADING NOTE Twice the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

1433

QUESTION ID

S34-B3-Q4

TYPE

Short answer

CORRECT ANSWER

All its vertices lie on the circle

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

True or false?

An inscribed triangle can have a vertex inside the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

1434

QUESTION ID

S34-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All vertices are on the circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

1436

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

AC = CE = EA

GRADING NOTE REQUIRED WORK: Show each central angle between consecutive marks is 60°, so AOC, COE, EOA are each 120°, then use congruent triangles (SAS).

Any valid proof. Essential: each small triangle (O, two consecutive marks) is equilateral, so each central angle is 60°; ∠AOC = ∠COE = ∠EOA = 120°; triangles AOC, COE, EOA are congruent by SAS (radii); so AC = CE = EA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has center O and diameter AD. An arc centered at D with radius DO meets the circle at B and C. Prove that triangle ABC is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

1437

QUESTION ID

S34-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

AB = BC = CA

GRADING NOTE REQUIRED WORK: Show △DOB and △DOC are equilateral, find the central angles, and use SAS (or inscribed angles).

Any valid proof. Essential: DB = DO = OB, so △DOB is equilateral and ∠DOB = 60°; likewise ∠DOC = 60°; so ∠BOC = 120° and ∠AOB = ∠AOC = 180° − 60° = 120°; SAS with radii gives AB = BC = CA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why stepping a compass set to the radius around a circle returns exactly to the starting point after six steps.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

1438

QUESTION ID

S34-F1-Q3

TYPE

Answer with required work

CORRECT ANSWER

Each step makes a 60° central angle, and 6 × 60° = 360°.

GRADING NOTE REQUIRED WORK: Use the equilateral triangles formed with the center.

Any valid explanation. Essential: each chord equals the radius, so the triangle with the center is equilateral (60° at O); six steps fill 360°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the stepped construction, points A, B, C, D, E, F lie on a circle with center O, each 60° from the next. Prove that ∠CAE = 60°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

1439

QUESTION ID

S34-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

m∠CAE = 60°

GRADING NOTE REQUIRED WORK: Use the inscribed angle theorem (or the isosceles triangles AOC and AOE).

Any valid proof. Essential: arc CE (not containing A) measures 120°, and an inscribed angle is half its arc. Alternative: ∠OAC = ∠OAE = 30° from isosceles triangles with 120° apex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Ana inscribes a triangle by marking points every 120° around the center with a protractor. Explain why her triangle is equilateral, and why her method is not a compass-and-straightedge construction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

1440

QUESTION ID

S34-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Equal central angles of 120° cut off equal chords (SAS with radii), so the sides are equal. Measuring with a protractor is not one of the allowed construction tools.

GRADING NOTE Any equivalent explanation with both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

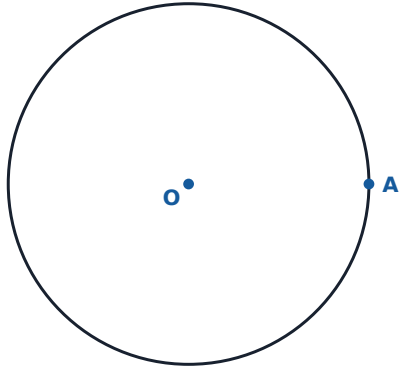
SET 34 • HSG-CO.D.13

FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

1442

QUESTION ID

S34-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Mark six points by stepping the radius, then bisect each of the six arcs (perpendicular bisector of each chord) to get six more; join the 12 points in order.

GRADING NOTE Grade the construction marks, not only the final figure: the six stepped points and the bisector constructions for the arcs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

1443

QUESTION ID
S34-F2-Q2

TYPE
Short answer

CORRECT ANSWER
30°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

True or false?

Bisecting the arcs of an inscribed regular hexagon gives the vertices of a regular nonagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

1444

QUESTION ID

S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Doubling the vertices gives a regular 12-gon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

A. pentagon

B. hexagon

C. decagon

D. octagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

1445

QUESTION ID

S34-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. octagon

GRADING NOTE 4 → 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

1446

QUESTION ID

S34-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 3 → 6 → 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a square inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct a perpendicular bisector	**HSG-CO.D.12**
BACKWARD 2	Diagonals of a square	**HSG-CO.C.11**
BACKWARD 3	Diameters and the center of a circle	**7.G.B.4**
FORWARD 1	Justify that the inscribed quadrilateral is a square	**HSG-CO.C.11**
FORWARD 2	Construct a regular octagon inscribed in a circle	**HSG-CO.D.13**

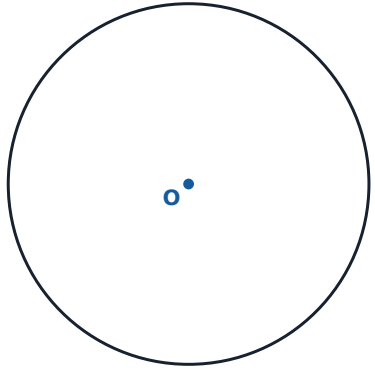
SET 35 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

1449

QUESTION ID

S35-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Draw a diameter AC through O; construct the perpendicular bisector of AC (a second diameter BD); join A, B, C, D.

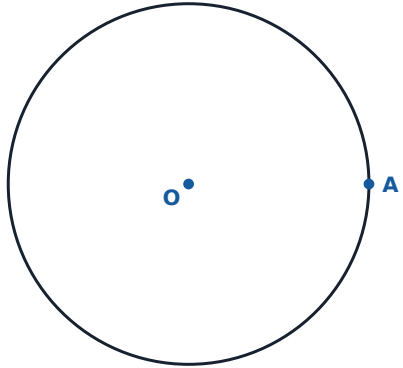
GRADING NOTE Grade the construction marks, not only the final figure: a diameter, the perpendicular bisector construction for it (arcs), and the square joining the four endpoints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

1450

QUESTION ID

S35-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Draw diameter AC through O; construct the perpendicular to AC at O, meeting the circle at B and D; join ABCD.

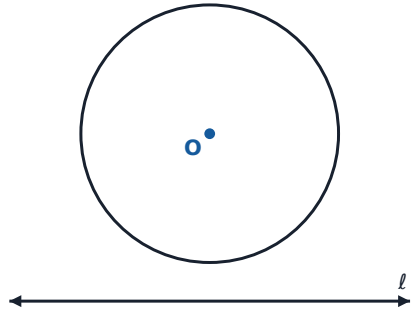
GRADING NOTE Grade the construction marks, not only the final figure: diameter through A and the perpendicular construction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line ℓ.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

1451

QUESTION ID

S35-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Construct the perpendicular from O to ℓ (one diameter); bisect the right angles at O to get diagonals at 45°; their endpoints form a square with sides parallel to ℓ.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular, the angle bisectors, and the square. Accept any valid method.

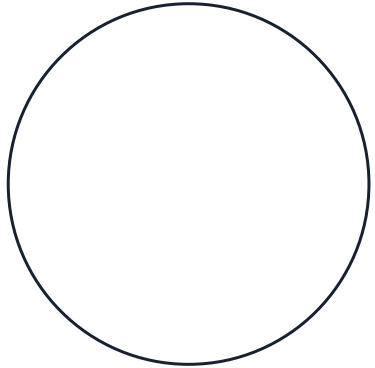
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a square inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

1452

QUESTION ID

S35-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Locate the center O with the perpendicular bisectors of two chords; draw a diameter through O; construct the perpendicular diameter; join the four endpoints.

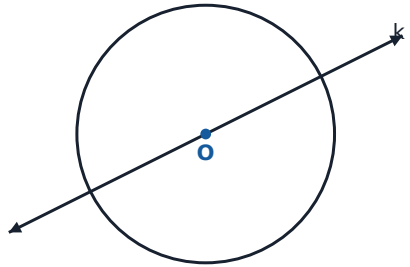
GRADING NOTE Grade the construction marks, not only the final figure: the center construction, two perpendicular diameters, and the square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle so that one of its diagonals lies along line k.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

1453

QUESTION ID

S35-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Line k meets the circle at A and C; construct the perpendicular to k at O, meeting the circle at B and D; join ABCD.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular construction at O and the square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

1455

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The perpendicular bisector of the diameter

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of a diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

1456

QUESTION ID

S35-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The center is the diameter's midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

A. six 60° arcs

B. four 45° arcs

C. four 90° arcs

D. two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

1457

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. four 90° arcs

GRADING NOTE 360 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

1458

QUESTION ID

S35-B1-Q4

TYPE

Short answer

CORRECT ANSWER

At the center, 5 cm from each end

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A perpendicular bisector can be drawn by measuring with a ruler only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

1459

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A construction uses compass and straightedge.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

The diagonals of a square are congruent and perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

1461

QUESTION ID

S35-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A square is a rectangle and a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. What is its most specific name?

A. rhombus that is not a square

B. square

C. trapezoid

D. kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

1462

QUESTION ID

S35-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. square

GRADING NOTE All three properties together give a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

1463

QUESTION ID

S35-B2-Q3

TYPE

Short answer

CORRECT ANSWER

4√2

GRADING NOTE Form required: exact value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

True or false?

A rectangle always has perpendicular diagonals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

1464

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only a square does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

1465

QUESTION ID

S35-B2-Q5

TYPE

Short answer

CORRECT ANSWER

90°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

1467

QUESTION ID

S35-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

True or false?

Every diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

1468

QUESTION ID

S35-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

A. At the midpoint of a radius

B. On the circle

C. At the center

D. Nowhere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

1469

QUESTION ID

S35-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. At the center

GRADING NOTE Both pass through the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

1470

QUESTION ID

S35-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A diameter

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

True or false?

All diameters of a circle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

1471

QUESTION ID

S35-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE All equal 2r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

1473

QUESTION ID

S35-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

ABCD is a square.

GRADING NOTE

REQUIRED WORK: Use that the diagonals are congruent (both diameters), bisect each other (radii), and are perpendicular; or SAS on the four triangles.

Any valid proof. Essential: OA = OB = OC = OD; four right angles at O; SAS gives four congruent sides; each triangle is isosceles right, so each vertex angle is 45° + 45° = 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

1474

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Side $5\sqrt{2}$; area 50

GRADING NOTE Form required: exact side and area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

True or false?

The side of a square inscribed in a circle of radius r is 2r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

1475

QUESTION ID

S35-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The side is r√2; 2r is its diagonal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

The area of a square inscribed in a circle of radius r is

A. $2r^2$

B. πr^2

C. $4r^2$

D. r^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

1476

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $2r^2$

GRADING NOTE $(r\sqrt{2})^2 = 2r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

1477

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2/π

GRADING NOTE Form required: exact value. $2r^2 \div \pi r^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

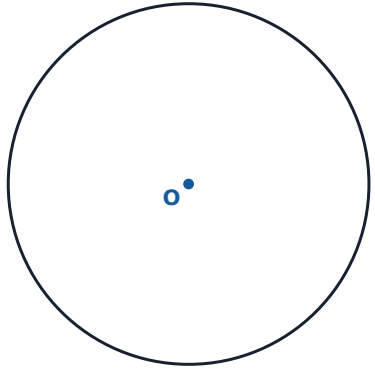
SET 35 • HSG-CO.D.13

FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

1479

QUESTION ID

S35-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Construct two perpendicular diameters, then bisect two adjacent right angles at O (or the four arcs); the eight points on the circle are the vertices.

GRADING NOTE Grade the construction marks, not only the final figure: the perpendicular diameters and the bisector constructions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

1480

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

45°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

True or false?

A regular octagon has interior angles of 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

1481

QUESTION ID

S35-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (8 − 2) · 180 ÷ 8 = 135°; 120° is the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

A. hexagon

B. 16-gon

C. 12-gon

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

1482

QUESTION ID

S35-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16-gon

GRADING NOTE 8 → 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

A regular octagon is constructed in a circle by bisecting the four 90° central angles of an inscribed square. Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

1483

QUESTION ID

S35-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Each central angle is half of 90°, which is 45°, and equal central angles give equal chords.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a regular hexagon inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct an inscribed equilateral triangle	**HSG-CO.D.13**
BACKWARD 2	An equilateral triangle has equal sides and 60° angles	**4.G.A.2**
BACKWARD 3	Circle facts: radius and chord	**7.G.B.4**
FORWARD 1	Justify that the inscribed hexagon is regular	**HSG-C.A.2**
FORWARD 2	Use inscribed hexagons to estimate π	**HSG-GMD.A.1**

SET 36 • HSG-CO.D.13

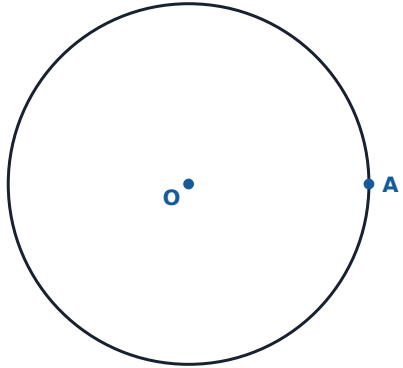
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

1486

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Set the compass to the radius OA; from A step around the circle marking 5 more points; join the six points in order.

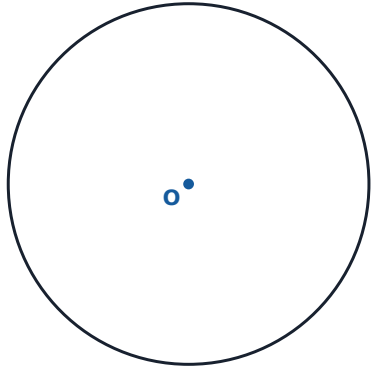
GRADING NOTE Grade the construction marks, not only the final figure: radius-length arcs around the circle and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

1487

QUESTION ID

S36-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Mark a point, step the radius around the circle six times, and join consecutive points.

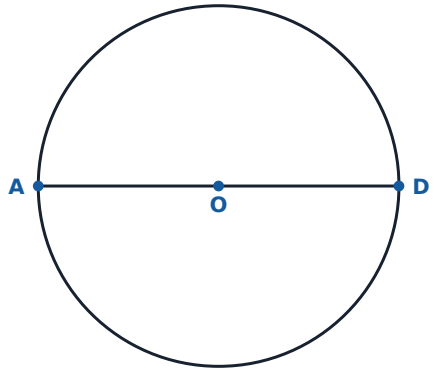
GRADING NOTE Grade the construction marks, not only the final figure: all stepping arcs and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

1488

QUESTION ID

S36-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

The arc at A meets the circle at B and F; the arc at D meets it at C and E; join A, B, C, D, E, F.

GRADING NOTE Grade the construction marks, not only the final figure: the two arcs and the hexagon.

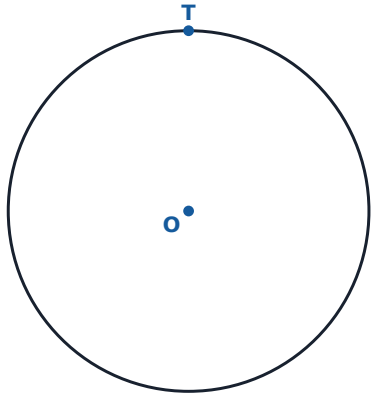
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle with one vertex at the top point T.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

1489

QUESTION ID

S36-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Keep the compass set to the radius and step around the circle from T; join the six points in order.

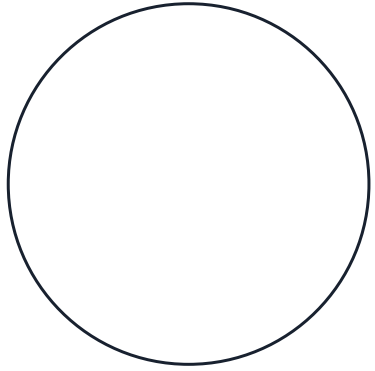
GRADING NOTE Grade the construction marks, not only the final figure: all stepping arcs from T and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a regular hexagon inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

1490

QUESTION ID

S36-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Locate the center with the perpendicular bisectors of two chords; set the compass to the radius; step around the circle and join consecutive points.

GRADING NOTE Grade the construction marks, not only the final figure: the center construction, the radius setting, and the hexagon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

ANSWER KEY Reduced copy of the question page, with its answer

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

1492

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The triangle joins every other vertex of the hexagon.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

True or false?

Both constructions start by stepping the radius around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

1493

QUESTION ID

S36-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same six points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

A. 12

B. 6

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

1494

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE 60° each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

1495

QUESTION ID

S36-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

True or false?

The inscribed equilateral triangle and hexagon have the same side length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

1496

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Triangle side r√3; hexagon side r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

True or false?

Each angle of an equilateral triangle measures 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

1498

QUESTION ID

S36-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 180 ÷ 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

1499

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Equilateral

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Six equilateral triangles meeting at a point fill an angle of

A. 300°

B. 360°

C. 60°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1500

QUESTION ID
S36-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 360°

GRADING NOTE 6 × 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

1500

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

1501

QUESTION ID

S36-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon splits into four equilateral triangles from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

1502

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Six.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

1504

QUESTION ID

S36-B3-Q1

TYPE

Short answer

CORRECT ANSWER

60°

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

True or false?

A chord equal to the radius forms an equilateral triangle with the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

1505

GRADING NOTE All sides equal r.

QUESTION ID

S36-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

A. 21

B. 14

C. 42

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

1506

QUESTION ID

S36-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42

GRADING NOTE 6 × 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

1507

QUESTION ID

S36-B3-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

True or false?

Every chord of a circle is as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

1508

QUESTION ID

S36-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Chords range up to the diameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

1510

QUESTION ID

S36-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

The hexagon is regular.

GRADING NOTE REQUIRED WORK: Show the six triangles from O are equilateral (all sides r), so all sides equal r and all angles equal 120°.

Any valid proof. Essential: each triangle O-consecutive points has three sides equal to r (equilateral, 60° at O); six 60° angles fill 360°, so the stepping closes; all sides equal r; each hexagon angle is 60° + 60° = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

A compass set to the radius of a circle is stepped around the circle from a starting point. Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

1511

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Each step makes a 60° central angle, and 6 × 60° = 360°.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Each interior angle of a regular hexagon is 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

1512

QUESTION ID

S36-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each interior angle is 120°; 60° is the central angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

A. 12

B. 48

C. 24

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

1513

QUESTION ID

S36-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 24

GRADING NOTE 6 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

1514

QUESTION ID

S36-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$6\sqrt{3}$

GRADING NOTE Form required: exact. Six equilateral triangles of side 2: $6 \cdot \sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

1516

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Perimeter

6r < circumference

2πr, so π > 3.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.212. What estimate of π does it give, to the nearest hundredth?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

1517

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 3.11

(π > 3.10)

GRADING NOTE Rounding to 0.01 required, with the explanation (half the perimeter: 6.212 ÷ 2 = 3.106).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

True or false?

Doubling the number of sides of an inscribed regular polygon makes its perimeter longer than the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

1518

QUESTION ID

S36-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter gets closer to the circumference but stays shorter (each side is a chord).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

A. Inscribed regular 96-gon

B. Inscribed regular hexagon

C. Inscribed square

D. Inscribed triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

1519

QUESTION ID

S36-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Inscribed regular 96-gon

GRADING NOTE More sides fit the circle more closely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

1520

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Each side is a chord, shorter than the arc it cuts off.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

Verify that a dilation maps a line to a parallel line (or to itself through the center)

MAIN	Geometry main question	**HSG-SRT.A.1.a**
BACKWARD 1	Dilate points with coordinates	**8.G.A.3**
BACKWARD 2	Compare slopes to test whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Dilate lines about a center other than the origin	**HSG-SRT.A.1.a**
FORWARD 2	Prove algebraically that a dilation maps a line to a parallel line	**HSG-SRT.A.1.a**

SET 37 • HSG-SRT.A.1.a

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is $y = 2x + 1$ (it does not pass through the origin). Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), write an equation of the image line, and state what this example shows about dilations and lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

1523

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 3) and (3, 9); y = 2x + 3; the image is parallel to ℓ (both slopes 2). This is evidence (one example) that a line not through the center maps to a parallel line.

GRADING NOTE Form required: both images, the equation, the comparison, and a statement that the example supports (does not prove) the property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is y = −x, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of (1, −1) and (3, −3). What is the image line, and what does this example suggest about lines through the center?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

1524

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −2) and (6, −6); the image is y = −x, the same line. It suggests that a line through the center maps onto itself.

GRADING NOTE Form required: the images, the image line, and the observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The horizontal line y = 4 is dilated about the center C(0, 2) with scale factor ½. Find the images of (0, 4) and (6, 4), write an equation of the image line, and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

1525

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

Images (0, 3) and (3, 3); image line y = 3; it is parallel to y = 4 (both horizontal), and y = 4 does not pass through C.

GRADING NOTE Form required: both images (center + ½ · (point − center)), the equation, and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

Line n is y = −½x + 4. Dilate it about the center C(2, 1) with scale factor 3: find the images of (0, 4) and (4, 2), write an equation of the image line, and compare it with n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

1526

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

Images (−4, 10) and (8, 4); image line y = −½x + 8; it is parallel to n (same slope −½), and n does not pass through C.

GRADING NOTE Form required: both images (center + 3 · (point − center)), the equation, and the comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q4**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

Line p is y = 2x, which passes through C(1, 2). Dilate it about C with scale factor 4: find the images of (0, 0) and (2, 4) and the image line. What do you observe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

1527

QUESTION ID

S37-M-Q5

TYPE

Short answer

CORRECT ANSWER

Images (−3, −6) and (5, 10); the image line is y = 2x, the same line, because p passes through the center.

GRADING NOTE Form required: both images, the image line, and the observation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (2, −3) about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

1529

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(8, −12)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

1530

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

A. 6

B. 9

C. 3

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

1531

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Each coordinate is tripled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

A dilation about the origin with scale factor 1 leaves every point where it is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

1532

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 moves the center to a new point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

1533

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The center is fixed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

1535

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

True or false?

y = 2x + 1 and y = 2x + 3 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

1536

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same slope, different intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to y = −3x + 2?

A. y = ⅓x

B. y = −3

C. y = −3x − 5

D. y = 3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

1537

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = −3x − 5

GRADING NOTE Slope −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

1538

QUESTION ID

S37-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Yes (both horizontal)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Lines with slopes 2 and −2 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

1539

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

1541

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y = 2x + 3

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

1542

QUESTION ID

S37-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y = −x

GRADING NOTE Any equivalent equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

Which point is on y = 2x + 3?

A. (5, 1)

B. (2, 3)

C. (0, 2)

D. (1, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

1543

QUESTION ID

S37-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 5)

GRADING NOTE 2(1) + 3 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

True or false?

The line through (0, 0) and (4, 8) is y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

1544

QUESTION ID

S37-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Slope 2, intercept 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

True or false?

The line through (1, 1) and (2, 4) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

1545

QUESTION ID

S37-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4 - 1) \div (2 - 1) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Dilate the line y = x + 2 about the center (2, 0) with scale factor 2. Find the images of (0, 2) and (2, 4) and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

1547

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 4) and (2, 8); y = x + 6, parallel to y = x + 2.

GRADING NOTE Form required: images, equation, and conclusion. Image = center + 2(point − center).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does the line $y = x - 2$ pass through the center $(2, 0)$? What happens to it under any dilation about $(2, 0)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

1548

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes, $0 = 2 - 2$; it maps onto itself.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

True or false?

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

1549

QUESTION ID

S37-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1, 1) + 3(1, 0) = (4, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

A. (4, 4)

B. (8, 10)

C. (2, 3)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

1550

QUESTION ID

S37-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 4)

GRADING NOTE (0, 2) + ½(4, 4) = (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ: $y = 3x$. A dilation about (1, 0) with scale factor 2 maps ℓ to ℓ'. Find ℓ'.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

1551

QUESTION ID

S37-F1-Q5

TYPE

Short answer

CORRECT ANSWER

y = 3x + 3

GRADING NOTE Any equivalent equation. (0, 0) → (−1, 0) and (1, 3) → (1, 6); slope 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the dilation about the origin with scale factor k ($k > 0$, $k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

1553

QUESTION ID

S37-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

The image of (x, mx + b) is (kx, kmx + kb), which satisfies y = mx + kb.

GRADING NOTE REQUIRED WORK: Take any point (x, mx + b) on the line, find its image, and show the image satisfies y = mx + kb (and every point of that line is an image).

Any valid proof (paragraph, two-column, or flow). Essential: image (X, Y) = (kx, k(mx + b)); Y = m(kx) + kb = mX + kb; conversely (X, mX + kb) is the image of (X/k, mX/k + b). Conclusion: same slope m, so parallel when b ≠ 0, the same line when b = 0. Invalid: checking only two points of one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

1554

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

When b = 0 (the line passes through the center).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

True or false?

The dilation $(x, y) \to (kx, ky)$ maps the vertical line $x = c$ to the line $x = c + k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

1555

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It maps $x = c$ to $x = kc$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

A. $y = -8x + 5$

B. $y = -2x + 20$

C. $y = -2x + 5$

D. $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

1556

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $y = -2x + 20$

GRADING NOTE $y = mx + kb$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

1557

QUESTION ID
S37-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Examples cannot cover every line; a proof must work for all lines (with variables).

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

Verify that a dilation multiplies lengths by the scale factor

MAIN	Geometry main question	**HSG-SRT.A.1.b**
BACKWARD 1	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 2	Scale factor in scale drawings	**7.G.A.1**
BACKWARD 3	Dilate points about the origin	**8.G.A.3**
FORWARD 1	Find the center or scale factor from a segment and its image	**HSG-SRT.A.1.b**
FORWARD 2	Perimeter and area of dilated figures	**HSG-SRT.A.2**

SET 38 • HSG-SRT.A.1.b

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

1560

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

A′(3, 6), B′(12, 18); AB = 5 and A′B′ = 15 = 3 · AB.

GRADING NOTE Form required: images, both lengths, and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

1561

QUESTION ID
S38-M-Q2

TYPE
Short answer

CORRECT ANSWER
P′(−2, 1), Q′(4, −3); PQ = √208 = 4√13 ≈ 14.42 and P′Q′ = √52 = 2√13 ≈ 7.21, half as long.

GRADING NOTE Form required: both lengths (exact or to 0.01) and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

1562

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

C′ = (2, 1), D′ = (8, 9); C′D′ = 10 = 2 · CD (CD = 5).

GRADING NOTE Form required: images and lengths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q3**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Dilate it about (1, 1) with scale factor 2. Find all three image side lengths, compare each with the original, and say what the comparison supports.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

1563

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

A′(−1, −1), B′(7, −1), C′(−1, 5); A′B′ = 8 = 2 · 4, A′C′ = 6 = 2 · 3, B′C′ = 10 = 2 · 5. Every length doubled: evidence that a dilation multiplies lengths by k.

GRADING NOTE Form required: images, three lengths with ratios, and the conclusion stated as evidence (not a general proof).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

Segment EF has E(−3, −1) and F(3, 7). Dilate it about the origin with scale factor 0.4. Find EF and E′F′, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

1564

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

E′(−1.2, −0.4), F′(1.2, 2.8); EF = 10 and E′F′ = 4 = 0.4 · 10.

GRADING NOTE Form required: images, both lengths, and the ratio.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

1566

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

Find the distance between (−2, 1) and (4, −3).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

1567

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

√52 = 2√13

GRADING NOTE Any equivalent exact form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

A. √17

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

1568

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 13

GRADING NOTE √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

True or false?

The distance between (1, 2) and (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

1569

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √(9 + 16).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

1570

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

1572

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

1573

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5/1500 = 1/300

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

True or false?

A scale factor less than 1 makes a figure smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

1574

QUESTION ID

S38-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lengths shrink.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

A. 5.5 by 7.5

B. 4 by 9

C. 8 by 12

D. 6 by 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

1575

QUESTION ID

S38-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6 by 9

GRADING NOTE Multiply each side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

Enlarging by a scale factor of 2 doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

1576

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Area is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

1578

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(12, 18)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

1579

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 1)

GRADING NOTE Form required: an ordered pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

A. 2.5

B. 3

C. 0.4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

1580

GRADING NOTE 5 ÷ 2.

QUESTION ID

S38-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

True or false?

Dilating about the origin multiplies both coordinates by the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

1581

QUESTION ID

S38-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (kx, ky).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

True or false?

Dilating (3, 0) by 2 about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

1582

QUESTION ID

S38-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (6, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

1584

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q1**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

1585

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Scale factor 2; center (−1, −1).

GRADING NOTE Form required: both. A′B′ = 2 = 2AB; lines AA′ and BB′ meet at (−1, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

True or false?

Under a dilation with scale factor k ≠ 1, the line through any point P other than the center and its image P′ passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

1586

QUESTION ID

S38-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P′ is on ray CP from the center C, and P′ ≠ P when P ≠ C and k ≠ 1, so line PP′ is line CP.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

A. 4

B. −3

C. ¼

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

1587

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. ¼

GRADING NOTE 1 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

1588

QUESTION ID
S38-F1-Q5

TYPE
Short answer

CORRECT ANSWER
$\frac{2}{3}$

GRADING NOTE Any equivalent form. 9k = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

1590

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Perimeter 54; area 108

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

1591

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{5}$

GRADING NOTE Any equivalent form. $k^2 = 8/50 = 4/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

True or false?

If lengths are multiplied by k, areas are multiplied by 2k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

1592

QUESTION ID
S38-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Areas are multiplied by k^2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

A. ½

B. 4

C. 2

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

1593

QUESTION ID

S38-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ¼

GRADING NOTE $(½)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

1594

QUESTION ID

S38-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Every side is multiplied by k, and the perimeter is the sum of the sides.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

Decide whether two figures are similar using similarity transformations

MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Similar figures as a sequence of transformations including a dilati…	**8.G.A.4**
BACKWARD 2	Specify a sequence of rigid motions	**HSG-CO.A.5**
BACKWARD 3	Dilate figures about the origin	**8.G.A.3**
FORWARD 1	Find the scale factor and correspondence when the image is turne	**HSG-SRT.A.2**
FORWARD 2	Prove that all circles are similar	**HSG-C.A.1**

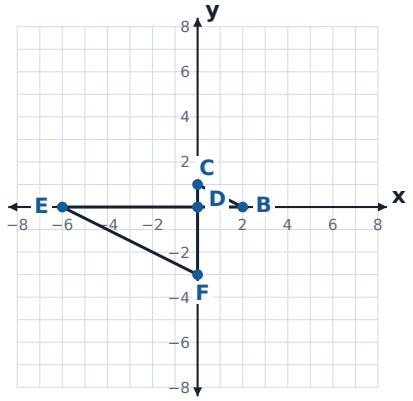
SET 39 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

1597

QUESTION ID
S39-M-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. Dilate by 3 about the origin, then rotate 180° about the origin.

GRADING NOTE Any valid similarity transformation. Check: (2, 0) → (6, 0) → (−6, 0); (0, 1) → (0, 3) → (0, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

1598

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. A dilation would need factor 6/4 = 1.5 for one side and 8/6 ≈ 1.33 for the other; no single scale factor works.

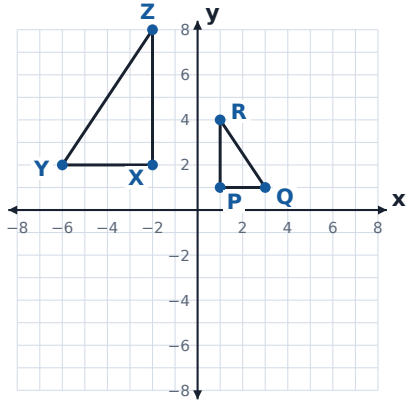
GRADING NOTE Any equivalent explanation comparing the ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

1599

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes. Dilate by 2 about the origin, then reflect across the y-axis.

GRADING NOTE Any valid sequence. (1, 1) → (2, 2) → (−2, 2); (3, 1) → (6, 2) → (−6, 2); (1, 4) → (2, 8) → (−2, 8).

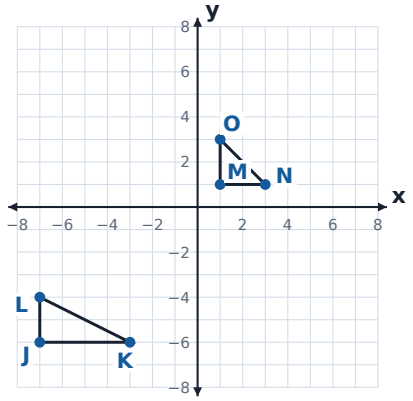
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Is triangle JKL similar to triangle MNO? If so, describe a similarity transformation; if not, explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5

1600

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. JKL has legs 4 and 2 (ratio 2 : 1) while MNO has legs 2 and 2 (ratio 1 : 1); a dilation multiplies both legs by the same factor, so no similarity transformation maps one onto the other.

GRADING NOTE Any equivalent explanation comparing ratios of corresponding sides (or the unequal acute angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q4/5